

SUPPLEMENT A TO "ON THE RATE OF CONVERGENCE OF FULLY CONNECTED DEEP NEURAL NETWORK REGRESSION ESTIMATES"

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This supplementary file contains the long and rather technical
proof of Theorem 2 and the induction proof of Theorem 3, that shows
the accuracy of the network.

A. APPENDIX: NETWORK APPROXIMATION OF SMOOTH FUNCTIONS.

A.1. *Proof of Theorem 2.* In this section we prove Theorem 2. The main idea is to construct the (p, C) -smooth function f by piecewise Taylor polynomials on a partition of cubes of $[-a, a]^d$. We first introduce some further notations.

A.1.1. *Notation.* Beside the notations of our main article we will use the following: If C is a cube we denote the "bottom left" corner of C by C_{left} . Therefore, each half-open cube C with side length s can be written as a polytope defined by

$$-x^{(j)} + C_{left}^{(j)} \leq 0 \text{ and } x^{(j)} - C_{left}^{(j)} - s < 0 \quad (j \in \{1, \dots, d\}).$$

Furthermore, we describe by $C_\delta^0 \subset C$ the cube, which contains all $\mathbf{x} \in C$ that lie with a distance of at least δ to the boundaries of C , i.e. a polytope defined by

$$-x^{(j)} + C_{left}^{(j)} \leq -\delta \text{ and } x^{(j)} - C_{left}^{(j)} - s < -\delta \quad (j \in \{1, \dots, d\}).$$

If $\mathcal{P}$ is a partition of cubes of $[-a, a]^d$ and $\mathbf{x} \in [-a, a]^d$, then we denote the cube $C \in \mathcal{P}$, which satisfies $\mathbf{x} \in C$, by $C_{\mathcal{P}}(\mathbf{x})$.

A.1.2. *An auxiliary result.* In the proof of Theorem 2 we will use the following lemma, which shows that every (p, C) -smooth function can be approximated by a Taylor polynomial.

LEMMA 1. *Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $s \in (0, 1]$, and let $C > 0$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function, let $\mathbf{x}_0 \in \mathbb{R}^d$ and let $T_{f,q,\mathbf{x}_0}$ be the Taylor polynomial of total degree q around $\mathbf{x}_0$ defined by*

$$T_{f,q,\mathbf{x}_0}(\mathbf{x}) = \sum_{j \in \mathbb{N}_0^d : \|\mathbf{j}\|_1 \leq q} (\partial^{\mathbf{j}} f)(\mathbf{x}_0) \cdot \frac{(\mathbf{x} - \mathbf{x}_0)^{\mathbf{j}}}{\mathbf{j}!}$$

Then for any $\mathbf{x} \in \mathbb{R}^d$

$$|f(\mathbf{x}) - T_{f,q,\mathbf{x}_0}(\mathbf{x})| \leq c_{32} \cdot C \cdot \|\mathbf{x} - \mathbf{x}_0\|^p$$

holds for a constant $c_{32} = c_{32}(q, d)$ depending only on q and d .

PROOF. See Lemma 1 in [1]. □

A.1.3. *Idea of the proof of Theorem 2 a).* In the proof of Theorem 2 a) we use Lemma 1 and approximate our function by a piecewise Taylor polynomial. To define this piecewise Taylor polynomial, we partition $[-a, a]^d$ into M^d and M^{2d} half-open equivolume cubes of the form

$$[\boldsymbol{\alpha}, \boldsymbol{\beta}) = [\boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}) \times \cdots \times [\boldsymbol{\alpha}^{(d)}, \boldsymbol{\beta}^{(d)}), \quad \boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^d,$$

respectively. Let

$$(15) \quad \mathcal{P}_1 = \{C_{k,1}\}_{k \in \{1, \dots, M^d\}} \text{ and } \mathcal{P}_2 = \{C_{j,2}\}_{j \in \{1, \dots, M^{2d}\}}$$

be the corresponding partitions. We denote for each $i \in \{1, \dots, M^d\}$ those cubes of $\mathcal{P}_2$ that are contained in $C_{i,1}$ by $\tilde{C}_{1,i}, \dots, \tilde{C}_{M^d,i}$. Here we order the cubes in such a way that

$$(16) \quad (\tilde{C}_{k,i})_{left} = (C_{i,1})_{left} + \mathbf{v}_k,$$

holds for all $k \in \{1, \dots, M^d\}, i \in \{1, \dots, M^d\}$ and for some vector $\mathbf{v}_k$ with entries in $\{0, 2a/M^2, \dots, (M-1) \cdot 2a/M^2\}$. The vector $\mathbf{v}_k$ describes the position of $(\tilde{C}_{k,i})_{left}$ relative to $(C_{i,1})_{left}$, and we order the above cubes such that this position is independent of i . Now it is easy to see, that the partition $\mathcal{P}_2$ can be represented by the cubes $\tilde{C}_{k,i}$. In particular, we have

$$\mathcal{P}_2 = \{\tilde{C}_{k,i}\}_{k \in \{1, \dots, M^d\}, i \in \{1, \dots, M^d\}}.$$

The Taylor expansion in Lemma 1 can then be computed by the piecewise Taylor polynomial defined on $\mathcal{P}_2$. In particular, we have

$$T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) = \sum_{k \in \{1, \dots, M^d\}, i \in \{1, \dots, M^d\}} T_{f,q,(\tilde{C}_{k,i})_{left}}(\mathbf{x}) \cdot \mathbb{1}_{\tilde{C}_{k,i}}(\mathbf{x})$$

and this piecewise Taylor polynomial satisfies

$$\sup_{\mathbf{x} \in [-a, a]^d} \left| f(\mathbf{x}) - T_{f, q, (C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) \right| \leq c_{32} \cdot (2 \cdot a \cdot d)^{2p} \cdot C \cdot \frac{1}{M^{2p}}.$$

To approximate $f(\mathbf{x})$ by neural networks our proof follows *four* key steps:

1. Compute $T_{f, q, (C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$ by using recursively defined functions.
2. Approximate the recursive functions by neural networks. The resulting network is a good approximation for $f(\mathbf{x})$ in case that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k, 2})_{1/M^{2p+2}}^0.$$

3. Construct a neural network to approximate $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$, where

$$w_{\mathcal{P}_2}(\mathbf{x}) = \prod_{j=1}^d \left(1 - \frac{M^2}{a} \cdot \left| (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} + \frac{a}{M^2} - x^{(j)} \right| \right)_+$$

is a linear tensorproduct B-spline which takes its maximum value at the center of $C_{\mathcal{P}_2}(\mathbf{x})$, which is nonzero in the inner part of $C_{\mathcal{P}_2}(\mathbf{x})$ and which vanishes outside of $C_{\mathcal{P}_2}(\mathbf{x})$.

4. Apply those networks to 2^d slightly shifted partitions of $\mathcal{P}_2$ to approximate $f(\mathbf{x})$ in supremum norm.

A.1.4. *Key step 1: A recursive definition of $T_{f, q, (C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$.* The following recursive definition of the piecewise Taylor polynomial will later help us to define a neural network approximating the function f . Let $i \in \{1, \dots, M^d\}$ and $C_{\mathcal{P}_1}(\mathbf{x}) = C_{i, 1}$. The recursion follows *two* steps. In a first step we compute the value of $(C_{\mathcal{P}_1}(\mathbf{x}))_{left} = (C_{i, 1})_{left}$ and the values of $(\partial^{\mathbf{l}} f)((\tilde{C}_{j, i})_{left})$ for $j \in \{1, \dots, M^d\}$ and $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. This can be done by computing the indicator function $\mathbb{1}_{C_{i, 1}}$ multiplied by $(C_{i, 1})_{left}$ or $(\partial^{\mathbf{l}} f)((\tilde{C}_{j, i})_{left})$ for each $i \in \{1, \dots, M^d\}$, respectively. Furthermore we need the value of the input $\mathbf{x}$ in the further recursive definition, such that we shift this value by applying the identity function. We set

$$\phi_{1, 1} = (\phi_{1, 1}^{(1)}, \dots, \phi_{1, 1}^{(d)}) = \mathbf{x},$$

$$\phi_{2, 1} = (\phi_{2, 1}^{(1)}, \dots, \phi_{2, 1}^{(d)}) = \sum_{i \in \{1, \dots, M^d\}} (C_{i, 1})_{left} \cdot \mathbb{1}_{C_{i, 1}}(\mathbf{x})$$

and

$$\phi_{3,1}^{(1,j)} = \sum_{i \in \{1, \dots, M^d\}} (\partial^{\mathbf{l}} f) \left((\tilde{C}_{j,i})_{left} \right) \cdot \mathbb{1}_{C_{i,1}}(\mathbf{x}),$$

for $j \in \{1, \dots, M^d\}$ and $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$.

Let $i, j \in \{1, \dots, M^d\}$ and $(C_{\mathcal{P}_2}(\mathbf{x}))_{left} = (\tilde{C}_{j,i})_{left}$. In a second step of the recursion we compute the value of $(C_{\mathcal{P}_2}(\mathbf{x}))_{left} = (\tilde{C}_{j,i})_{left}$ and the values of $(\partial^{\mathbf{l}} f)((C_{\mathcal{P}_2}(\mathbf{x}))_{left})$ for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. It is easy to see that each cube $\tilde{C}_{j,i}$ can be defined by

$$(17) \quad \mathcal{A}^{(j)} = \left\{ \mathbf{x} \in \mathbb{R}^d : -x^{(k)} + \phi_{2,1}^{(k)} + v_j^{(k)} \leq 0 \right. \\ \left. \text{and } x^{(k)} - \phi_{2,1}^{(k)} - v_j^{(k)} - \frac{2a}{M^2} < 0 \text{ for all } k \in \{1, \dots, d\} \right\}.$$

Thus in our recursion we compute for each $j \in \{1, \dots, M^d\}$ the indicator function $\mathbb{1}_{\mathcal{A}^{(j)}}$ multiplied by $\phi_{2,1} + \mathbf{v}_j$ or $\phi_{3,1}^{(1,j)}$ for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. Again we shift the value of $\mathbf{x}$ by applying the identity function. We set

$$\phi_{1,2} = (\phi_{1,2}^{(1)}, \dots, \phi_{1,2}^{(d)})^T = \phi_{1,1},$$

$$\phi_{2,2} = (\phi_{2,2}^{(1)}, \dots, \phi_{2,2}^{(d)}) = \sum_{j=1}^{M^d} (\phi_{2,1} + \mathbf{v}_j) \cdot \mathbb{1}_{\mathcal{A}^{(j)}}(\phi_{1,1})$$

and

$$\phi_{3,2}^{(1)} = \sum_{j=1}^{M^d} \phi_{3,1}^{(1,j)} \cdot \mathbb{1}_{\mathcal{A}^{(j)}}(\phi_{1,1})$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. In a last step we compute the Taylor polynomial by

$$\phi_{1,3} = \sum_{\mathbf{j} \in \mathbb{N}_0^d : \|\mathbf{j}\|_1 \leq q} \frac{\phi_{3,2}^{(\mathbf{j})}}{\mathbf{j}!} \cdot (\phi_{1,2} - \phi_{2,2})^{\mathbf{j}}.$$

Our next lemma shows that this recursion computes our piecewise Taylor polynomial.

LEMMA 2. *Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $s \in (0, 1]$, let $C > 0$ and $\mathbf{x} \in [-a, a]^d$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function and let $T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}$ be the Taylor polynomial of total degree q around $(C_{\mathcal{P}_2}(\mathbf{x}))_{left}$. Define $\phi_{1,3}$ recursively as above. Then we have*

$$\phi_{1,3} = T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}).$$

PROOF. Let $j, i \in \{1, \dots, M^d\}$ and $\mathbf{x} \in \tilde{C}_{j,i}$, which implies $C_{\mathcal{P}_2}(\mathbf{x}) = \tilde{C}_{j,i}$ and $\mathbf{x} \in C_{i,1}$. Now it is easy to see that

$$\phi_{2,1} = (C_{i,1})_{left}$$

and

$$\phi_{3,1}^{(1,j)} = (\partial^1 f)((\tilde{C}_{j,i})_{left}), \quad j \in \{1, \dots, M^d\}.$$

According to (16) we then have

$$\phi_{2,1} + \mathbf{v}_j = (\tilde{C}_{j,i})_{left}$$

and $\mathcal{A}^{(j)}$ describes the cube $\tilde{C}_{j,i}$. Together with $\phi_{1,1} = \mathbf{x}$ it follows, that

$$\phi_{2,2} = (\tilde{C}_{j,i})_{left}$$

and

$$\phi_{3,2}^{(\mathbf{l})} = (\partial^{\mathbf{l}} f)((\tilde{C}_{j,i})_{left}),$$

which implies

$$\phi_{1,3} = T_{f,q,(\tilde{C}_{j,i})_{left}}(\mathbf{x}) = T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}).$$

□

A.1.5. *Key step 2: Approximating $\phi_{1,3}$ by neural networks.* The basic idea of the proof is to define a composed neural network, which approximately computes the functions in the definition of $\phi_{1,1}$, $\phi_{2,1}$, $\phi_{3,1}^{(1,j)}$, $\phi_{1,2}$, $\phi_{2,2}$, $\phi_{3,2}^{(\mathbf{l})}$, $\phi_{1,3}$ for $j \in \{1, \dots, M^d\}$ and $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. We will show, that this neural network is a good approximation for $f(\mathbf{x})$ in case that $\mathbf{x}$ does not lie close to the boundary of any cube of $\mathcal{P}_2$, i.e. for

$$\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0.$$

LEMMA 3. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $\mathcal{P}_2$ be defined as in (15). Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $s \in (0, 1]$, and let $C > 0$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function. Let $1 \leq a < \infty$. Then there exists for $M \in \mathbb{N}$ sufficiently large (independent of the size of a , but

$$M^{2p} \geq c_{32} \cdot \max \left\{ \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)}, (2 \cdot a \cdot d)^{2p} \cdot C \right\}$$

must hold), a neural network $\hat{f}_{\mathcal{P}_2} \in \mathcal{F}(L, r)$ with

- (i) $L = 4 + \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil$
- (ii) $r = \max \left\{ \left(\binom{d+q}{d} + d \right) \cdot M^d \cdot 2 \cdot (2+2d) + 2d, 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\}$

such that

$$|\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - f(\mathbf{x})| \leq c_{33} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}$$

holds for all $\mathbf{x} \in \bigcup_{j \in \{1, \dots, M^{2d}\}} (C_{j,2})_{1/M^{2p+2}}^0$. The network value is bounded by

$$|\hat{f}_{\mathcal{P}_2}(\mathbf{x})| \leq 2 \cdot e^{2ad} \cdot \max \left\{ \|f\|_{C^q([-a, a]^d)}, 1 \right\}$$

for all $\mathbf{x} \in [-a, a]^d$.

Lemma 3 shows already that neural networks with depth $c_{34} \cdot \log_4(M)$ and width $c_{35} \cdot M^d$ approximate $f(\mathbf{x})$ with a rate of order $1/M^{2p}$ in case that $\mathbf{x} \in \bigcup_{j \in \{1, \dots, M^{2d}\}} (C_{j,2})_{1/M^{2p+2}}^0$. To proof this result we need some auxiliary networks, i.e. neural networks for simpler function classes.

Auxiliary neural networks. To shift the input value in the next hidden layer or to synchronize the number of hidden layers for two networks we use the network $\hat{f}_{id} : \mathbb{R} \rightarrow \mathbb{R}$,

$$\hat{f}_{id}(z) = \sigma(z) - \sigma(-z) = z, \quad z \in \mathbb{R}$$

and

$$\hat{f}_{id}(\mathbf{x}) = \left(\hat{f}_{id}(x^{(1)}), \dots, \hat{f}_{id}(x^{(d)}) \right) = \left(x^{(1)}, \dots, x^{(d)} \right), \quad \mathbf{x} \in \mathbb{R}^d.$$

Here we will use the abbreviations

$$\hat{f}_{id}^0(\mathbf{x}) = \mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^d$$

$$\hat{f}_{id}^{t+1}(\mathbf{x}) = \hat{f}_{id}(\hat{f}_{id}^t(\mathbf{x})) = \mathbf{x}, \quad t \in \mathbb{N}_0, \mathbf{x} \in \mathbb{R}^d.$$

The next lemma presents a network which returns approximately xy for given input x and y .

LEMMA 4. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Then for any $R \in \mathbb{N}$ and any $a \geq 1$ a neural network*

$$\hat{f}_{mult} \in \mathcal{F}(R, 18)$$

exists such that

$$|\hat{f}_{mult}(x, y) - x \cdot y| \leq 2 \cdot a^2 \cdot 4^{-R}$$

holds for all $x, y \in [-a, a]$.

PROOF. A similar result can be found in Lemma A.2 in the Supplement of [2]. A proof of our result is given in Supplement B. $\square$

Let $\mathcal{P}_N$ be the linear span of all monomials of the form

$$\prod_{k=1}^d \left(x^{(k)}\right)^{r_k}$$

for some $r_1, \dots, r_d \in \mathbb{N}_0$, $r_1 + \dots + r_d \leq N$. Then, $\mathcal{P}_N$ is a linear vector space of functions of dimension

$$\dim \mathcal{P}_N = \left| \left\{ (r_0, \dots, r_d) \in \mathbb{N}_0^{d+1} : r_0 + \dots + r_d = N \right\} \right| = \binom{d+N}{d}.$$

In the next lemma, we construct a neural network that approximates functions of the class $\mathcal{P}_N$ multiplied by an additional factor. This modified form of polynomials is later needed in the construction of our network of the main result.

LEMMA 5. *Let $m_1, \dots, m_{\binom{d+N}{d}}$ denote all monomials in $\mathcal{P}_N$ for some $N \in \mathbb{N}$. Let $r_1, \dots, r_{\binom{d+N}{d}} \in \mathbb{R}$, define*

$$p\left(\mathbf{x}, y_1, \dots, y_{\binom{d+N}{d}}\right) = \sum_{i=1}^{\binom{d+N}{d}} r_i \cdot y_i \cdot m_i(\mathbf{x}), \quad \mathbf{x} \in [-a, a]^d, y_i \in [-a, a]$$

and set $\bar{r}(p) = \max_{i \in \{1, \dots, \binom{d+N}{d}\}} |r_i|$. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Then for any $a \geq 1$ and

$$(18) \quad R \geq \log_4(2 \cdot 4^{2 \cdot (N+1)} \cdot a^{2 \cdot (N+1)})$$

a neural network

$$\hat{f}_p \in \mathcal{F}(L, r)$$

with $L = R \cdot \lceil \log_2(N+1) \rceil$ and $r = 18 \cdot (N+1) \cdot \binom{d+N}{d}$ exists, such that

$$\left| \hat{f}_p \left(\mathbf{x}, y_1, \dots, y_{\binom{d+N}{d}} \right) - p \left(\mathbf{x}, y_1, \dots, y_{\binom{d+N}{d}} \right) \right| \leq c_{36} \cdot \bar{r}(p) \cdot a^{4(N+1)} \cdot 4^{-R}$$

for all $\mathbf{x} \in [-a, a]^d$, $y_1, \dots, y_{\binom{d+N}{d}} \in [-a, a]$, where c_{36} depends on d and N .

PROOF. A similar result can be found in Lemma A.4 in the Supplement of [2]. For sake of completeness we provide a proof of our result in Supplement B. $\square$

In the next lemma, we describe how to build a network that approximates the multidimensional indicator function and the multidimensional indicator function multiplied by an additional factor. In those networks we use the advantage of ReLU activation function, that it is zero in case of negative input. For instance, we use that for $R \in \mathbb{N}$

$$\sigma(1 - R \cdot \sigma(x)) = \begin{cases} 0 & \text{for } x \geq \frac{1}{R} \\ 1 & \text{for } x \leq 0. \end{cases}$$

Analogously we have for any $|s| \leq R$ that

$$\sigma(\hat{f}_{id}(s) - R^2 \cdot \sigma(x)) + \sigma(-\hat{f}_{id}(s) - R^2 \cdot \sigma(x)) = \begin{cases} 0 & \text{for } x \geq \frac{1}{R} \\ s & \text{for } x \leq 0. \end{cases}$$

LEMMA 6. Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $R \in \mathbb{N}$. Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ with

$$b^{(i)} - a^{(i)} \geq \frac{2}{R} \text{ for all } i \in \{1, \dots, d\}$$

and let

$$K_{1/R} = \{\mathbf{x} \in \mathbb{R}^d : x^{(i)} \notin [a^{(i)}, a^{(i)} + 1/R) \cup (b^{(i)} - 1/R, b^{(i)}) \text{ for all } i \in \{1, \dots, d\}\}.$$

a) Then the network

$$\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \sigma \left(1 - R \cdot \sum_{i=1}^d \left(\sigma \left(a^{(i)} + \frac{1}{R} - x^{(i)} \right) \right) \right)$$

$$+\sigma\left(x^{(i)}-b^{(i)}+\frac{1}{R}\right)\right)\right)$$

of the class $\mathcal{F}(2, 2d)$ satisfies

$$\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})$$

for $\mathbf{x} \in K_{1/R}$ and

$$\left|\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) - \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})\right| \leq 1$$

for $\mathbf{x} \in \mathbb{R}^d$.

b) Let $|s| \leq R$. Then the network

$$\begin{aligned} \hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) = & \sigma\left(\hat{f}_{id}(s) - R^2 \cdot \sum_{i=1}^d \left(\sigma\left(a^{(i)} + \frac{1}{R} - x^{(i)}\right) \right. \right. \\ & \left. \left. + \sigma\left(x^{(i)} - b^{(i)} + \frac{1}{R}\right)\right)\right) \\ & - \sigma\left(-\hat{f}_{id}(s) - R^2 \cdot \sum_{i=1}^d \left(\sigma\left(a^{(i)} + \frac{1}{R} - x^{(i)}\right) \right. \right. \\ & \left. \left. + \sigma\left(x^{(i)} - b^{(i)} + \frac{1}{R}\right)\right)\right) \end{aligned}$$

of the class $\mathcal{F}(2, 2 \cdot (2d + 2))$ satisfies

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) = s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})$$

for $\mathbf{x} \in K_{1/R}$ and

$$\left|\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) - s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})\right| \leq |s|$$

for $\mathbf{x} \in \mathbb{R}^d$.

PROOF. a) For $\mathbf{x} \in [\mathbf{a} + 1/R \cdot \mathbf{1}, \mathbf{b} - 1/R \cdot \mathbf{1}]$ we have

$$a^{(i)} + \frac{1}{R} - x^{(i)} \leq 0 \text{ and } x^{(i)} - b^{(i)} + \frac{1}{R} \leq 0 \text{ for all } i \in \{1, \dots, d\},$$

which implies

$$\sum_{i=1}^d \left(\sigma\left(a^{(i)} + \frac{1}{R} - x^{(i)}\right) + \sigma\left(x^{(i)} - b^{(i)} + \frac{1}{R}\right)\right) = 0$$

and

$$\hat{f}_{ind,[\mathbf{a},\mathbf{b})}(\mathbf{x}) = \sigma(1 - 0) = 1 = \mathbb{1}_{[\mathbf{a},\mathbf{b})}(\mathbf{x}).$$

For $\mathbf{x} \notin [\mathbf{a}, \mathbf{b})$ we know that there is a $j \in \{1, \dots, d\}$ which satisfies

$$x^{(j)} \leq a^{(j)} \text{ or } x^{(j)} \geq b^{(j)}.$$

This leads to

$$1 - R \cdot \sum_{i=1}^d \left(\sigma \left(a^{(i)} + \frac{1}{R} - x^{(i)} \right) + \sigma \left(x^{(i)} - b^{(i)} + \frac{1}{R} \right) \right) \leq 0$$

and therefore we have

$$\hat{f}_{ind,[\mathbf{a},\mathbf{b})}(\mathbf{x}) = 0 = \mathbb{1}_{[\mathbf{a},\mathbf{b})}(\mathbf{x}).$$

For $\mathbf{x} \in \mathbb{R}^d$ we have

$$\mathbb{1}_{[\mathbf{a},\mathbf{b})}(\mathbf{x}) \in \{0, 1\}$$

and

$$0 \leq \hat{f}_{ind,[\mathbf{a},\mathbf{b})}(\mathbf{x}) \leq 1.$$

b) For $\mathbf{x} \in [\mathbf{a} + 1/R \cdot \mathbf{1}, \mathbf{b} - 1/R \cdot \mathbf{1}]$ we have

$$a^{(i)} + \frac{1}{R} - x^{(i)} \leq 0 \text{ and } x^{(i)} - b^{(i)} + \frac{1}{R} \leq 0 \text{ for all } i \in \{1, \dots, d\},$$

which implies

$$\sum_{i=1}^d \left(\sigma \left(a^{(i)} + \frac{1}{R} - x^{(i)} \right) + \sigma \left(x^{(i)} - b^{(i)} + \frac{1}{R} \right) \right) = 0$$

and

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) = \sigma(\hat{f}_{id}(s)) - \sigma(-\hat{f}_{id}(s)) = s = s \cdot \mathbb{1}_{[\mathbf{a},\mathbf{b})}(\mathbf{x}).$$

For $\mathbf{x} \notin [\mathbf{a}, \mathbf{b})$ we know that there is a $j \in \{1, \dots, d\}$ which satisfies

$$x^{(j)} \leq a^{(j)} \text{ or } x^{(j)} \geq b^{(j)}.$$

In case $0 \leq s \leq R$ this leads to

$$\hat{f}_{id}(s) - R^2 \cdot \sum_{i=1}^d \left(\sigma \left(a^{(i)} + \frac{1}{R} - x^{(i)} \right) + \sigma \left(x^{(i)} - b^{(i)} + \frac{1}{R} \right) \right) \leq 0$$

and therefore we have

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) = 0 = s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x}).$$

Similarly the assertion follows in case $-R \leq s < 0$.

For $\mathbf{x} \in \mathbb{R}^d$ and $s \geq 0$ we have

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) \in [0, s]$$

and

$$s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x}) \in \{0, s\},$$

which implies

$$|\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) - s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})| \leq |s|.$$

Similarly the assertion follows in case $\mathbf{x} \in \mathbb{R}^d$ and $s < 0$. $\square$

Using the networks of [Lemma 5](#) and [Lemma 6](#) helps us to construct the recursive functions of $\phi_{1,3}$.

PROOF OF [LEMMA 3](#). In a *first step of the proof* we describe how the recursively defined function $\phi_{1,3}$ of [Lemma 2](#) can be approximated by neural networks. In the construction we will use the network

$$\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]} \in \mathcal{F}(2, 2d)$$

of [Lemma 6](#), which approximates the indicator function $\mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})$ for some $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ and $B_M \in \mathbb{N}$ with

$$b^{(i)} - a^{(i)} \geq \frac{2}{B_M} \text{ for all } i \in \{1, \dots, d\}$$

and the network

$$\hat{f}_{test} \in \mathcal{F}(2, 2 \cdot (2d + 2))$$

of [Lemma 6](#), which approximates

$$s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x}).$$

Observe that for $B_M \in \mathbb{N}$, $|s| \leq B_M$ and

$$x^{(i)} \notin \left[a^{(i)}, a^{(i)} + \frac{1}{B_M} \right) \cup \left(b^{(i)} - \frac{1}{B_M}, b^{(i)} \right) \text{ for all } i \in \{1, \dots, d\}$$

we have

$$\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})$$

and

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s)(\mathbf{x}) = s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x}).$$

Here we treat B_M as R in [Lemma 6](#). For some vector $\mathbf{v} \in \mathbb{R}^d$ it follows

$$\mathbf{v} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \left(v^{(1)} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}), \dots, v^{(d)} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) \right).$$

To compute the final Taylor polynomial in $\phi_{1,3}$ we use the network

$$\hat{f}_p \in \mathcal{F} \left(B_{M,p} \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil, 18 \cdot (q+1) \cdot \binom{d+q}{d} \right)$$

from [Lemma 5](#) satisfying

$$\begin{aligned} & \left| \hat{f}_p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{q}} \right) - p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{q}} \right) \right| \\ (19) \quad & \leq c_{36} \cdot \bar{r}(p) \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot 4^{-B_{M,p}} \end{aligned}$$

for all $z^{(1)}, \dots, z^{(d)}, y_1, \dots, y_{\binom{d+q}{d}}$ contained in

$$\left[-\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\}, \max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right],$$

where $B_{M,p} \in \mathbb{N}$ satisfying

$$B_{M,p} \geq \log_4 \left(2 \cdot 4^{2 \cdot (q+1)} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{2 \cdot (q+1)} \right)$$

is properly chosen (cf., [\(18\)](#)). Here we treat $B_{M,p}$ as R in [Lemma 5](#). In case that $q = 0$ we use a polynomial of degree 1 where the r_i 's of all coefficients greater than zero are chosen as zero. That is why we changed $\log_2(q+1)$ to $\log_2(\max\{q+1, 2\})$ in the definition of L in [Lemma 5](#).

Each network of the recursion of $\phi_{1,3}$ is now computed by a neural network.

To compute the values of $\phi_{1,1}$, $\phi_{2,1}$ and $\phi_{3,1}^{(1,j)}$ we use for $j \in \{1, \dots, M^d\}$, $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and $i \in \{1, \dots, d\}$ the networks

$$\hat{\phi}_{1,1} = \left(\hat{\phi}_{1,1}^{(1)}, \dots, \hat{\phi}_{1,1}^{(d)} \right) = \hat{f}_{id}^2(\mathbf{x}),$$

$$\hat{\phi}_{2,1} = \left(\hat{\phi}_{2,1}^{(1)}, \dots, \hat{\phi}_{2,1}^{(d)} \right) = \sum_{i \in \{1, \dots, M^d\}} (C_{i,1})_{left} \cdot \hat{f}_{ind, C_{i,1}}(\mathbf{x})$$

and

$$\hat{\phi}_{3,1}^{(1,j)} = \sum_{i \in \{1, \dots, M^d\}} (\partial^{\mathbf{l}} f) \left((\tilde{C}_{j,i})_{left} \right) \cdot \hat{f}_{ind, C_{i,1}}(\mathbf{x}).$$

To compute $\phi_{1,2}$, $\phi_{2,2}$ and $\phi_{3,2}^{(\mathbf{l})}$ we use the networks

$$\hat{\phi}_{1,2} = \left(\hat{\phi}_{1,2}^{(1)}, \dots, \hat{\phi}_{1,2}^{(d)} \right) = \hat{f}_{id}^2(\hat{\phi}_{1,1}),$$

$$\hat{\phi}_{2,2}^{(i)} = \sum_{j=1}^{M^d} \hat{f}_{test} \left(\hat{\phi}_{1,1}, \hat{\phi}_{2,1} + \mathbf{v}_j, \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1}, \hat{\phi}_{2,1}^{(i)} + v_j^{(i)} \right)$$

for $i \in \{1, \dots, d\}$ and

$$\hat{\phi}_{2,2} = (\hat{\phi}_{2,2}^{(1)}, \dots, \hat{\phi}_{2,2}^{(d)})$$

and

$$(20) \quad \hat{\phi}_{3,2}^{(\mathbf{l})} = \sum_{j=1}^{M^d} \hat{f}_{test} \left(\hat{\phi}_{1,1}, \hat{\phi}_{2,1} + \mathbf{v}_j, \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1}, \hat{\phi}_{3,1}^{(1,j)} \right).$$

Choose $\mathbf{l}_1, \dots, \mathbf{l}_{\binom{d+q}{d}}$ such that

$$\left\{ \mathbf{l}_1, \dots, \mathbf{l}_{\binom{d+q}{d}} \right\} = \left\{ (s_1, \dots, s_d) \in \mathbb{N}_0^d : s_1 + \dots + s_d \leq q \right\}$$

holds. The value of $\phi_{1,3}$ can then be computed by

$$(21) \quad \hat{\phi}_{1,3} = \hat{f}_p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{d}} \right),$$

where

$$\mathbf{z} = \hat{\phi}_{1,2} - \hat{\phi}_{2,2}$$

and

$$y_v = \hat{\phi}_{3,2}^{(\mathbf{l}_v)}$$

for $v \in \left\{1, \dots, \binom{d+q}{d}\right\}$. The coefficients $r_1, \dots, r_{\binom{d+q}{d}}$ in [Lemma 5](#) are chosen as

$$r_i = \frac{1}{\mathbf{l}_i!}, \quad i \in \left\{1, \dots, \binom{d+q}{d}\right\}.$$

It is easy to see, that the network $\hat{\phi}_{1,3}$ forms a composed network, where the networks $\hat{\phi}_{1,1}, \hat{\phi}_{2,1}, \hat{\phi}_{3,1}^{(\mathbf{l}_v, 1)}, \dots, \hat{\phi}_{3,1}^{(\mathbf{l}_v, M^d)}$ and the networks $\hat{\phi}_{1,2}, \hat{\phi}_{2,2}, \hat{\phi}_{3,2}^{(\mathbf{l}_v)}$ ($v \in \{1, \dots, \binom{d+q}{d}\}$) are computed in parallel (i.e., in the same layers), respectively. Thus we can conclude, that

$$(\hat{\phi}_{1,1}, \hat{\phi}_{2,1}, \hat{\phi}_{3,1}^{(\mathbf{l}_v, 1)}, \dots, \hat{\phi}_{3,1}^{(\mathbf{l}_v, M^d)})$$

needs $L_1 = 2$ hidden layers and $r_1 = 2d + d \cdot M^d \cdot 2d + M^d \cdot \binom{d+q}{d} \cdot 2d$ neurons per layer in total. Fig.1 illustrates the described computation in an acyclic graph. Here one sees why we only need $c_{37} \cdot M^d$ instead of $c_{38} \cdot M^{2d}$ many neurons per layer to compute this network. In particular, we use that neural networks with width M^d have M^{2d} many connections between two neighboring layers. Thus every derivative of f for every cube of $\mathcal{P}_2$ can be computed by choosing the derivatives as the weights in our network. Furthermore we can conclude

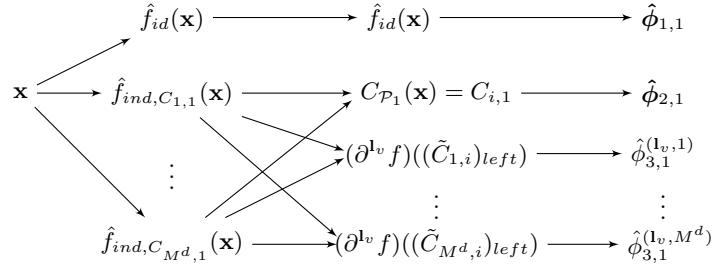


FIG 1. Computation of $(\hat{\phi}_{1,1}, \hat{\phi}_{2,1}, \hat{\phi}_{3,1}^{(\mathbf{l}_v, 1)}, \dots, \hat{\phi}_{3,1}^{(\mathbf{l}_v, M^d)})$

that

$$(\hat{\phi}_{1,2}, \hat{\phi}_{2,2}, \hat{\phi}_{3,2}^{(\mathbf{l}_v)})$$

needs $L_2 = L_1 + 2 = 4$ hidden layers and

$$\begin{aligned} r_2 &= \max\{r_1, 2d + d \cdot M^d \cdot 2 \cdot (2d + 2) + \binom{d+q}{d} \cdot M^d \cdot 2 \cdot (2d + 2)\} \\ &= 2d + \left(d + \binom{d+q}{d}\right) \cdot M^d \cdot 2 \cdot (2d + 2). \end{aligned}$$

neurons per layer. Finally we have that $\hat{\phi}_{1,3}$ lies in the class

$$\mathcal{F}(4 + B_{M,p} \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil, r)$$

with

$$r = \max \left\{ r_2, 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\}.$$

Here we have used, that

$$\mathcal{F}(L, r') \subseteq \mathcal{F}(L, r)$$

for $r' \leq r$. We set

$$\hat{f}_{\mathcal{P}_2}(\mathbf{x}) = \hat{\phi}_{1,3}.$$

In a *second step of the proof* we analyze the error of the network $\hat{f}_{\mathcal{P}_2}$ in case that

$$B_M \geq M^{2p+2}$$

and

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

From [Lemma 6](#) we can conclude that the networks $\hat{\phi}_{1,1}, \hat{\phi}_{2,1}, \hat{\phi}_{3,1}^{(\mathbf{l}_v, 1)}, \dots, \hat{\phi}_{3,1}^{(\mathbf{l}_v, M^d)}$ and the networks $\hat{\phi}_{1,2}, \hat{\phi}_{2,2}, \hat{\phi}_{3,2}^{(\mathbf{l}_v)}$ ($v \in \{1, \dots, \binom{d+q}{d}\}$) compute the corresponding functions $\phi_{1,1}, \phi_{2,1}, \phi_{3,1}^{(\mathbf{l}_v, 1)}, \dots, \phi_{3,1}^{(\mathbf{l}_v, M^d)}$ and $\phi_{1,2}, \phi_{2,2}, \phi_{3,2}^{(\mathbf{l}_v)}$ ($v \in \{1, \dots, \binom{d+q}{d}\}$) without an error. Thus it follows that

$$\left| \hat{\phi}_{1,2} - \hat{\phi}_{2,2} \right| = |\mathbf{x} - \phi_{2,2}| \leq 2a$$

and

$$\left| \hat{\phi}_{3,2}^{(\mathbf{l}_v)} \right| = \left| \phi_{3,2}^{(\mathbf{l}_v)} \right| \leq \|f\|_{C^q([-a,a]^d)}.$$

Therefore the input of $\hat{f}_p$ in [\(21\)](#) is contained in the interval where [\(19\)](#) holds. By choosing

$$B_{M,p} = \lceil \log_4(M^{2p}) \rceil$$

we get

$$\begin{aligned} \left| \hat{f}_{\mathcal{P}_2}(\mathbf{x}) - T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) \right| &= \left| \hat{\phi}_{1,3} - \phi_{1,3} \right| \\ &\leq c_{36} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a,a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}, \end{aligned}$$

where we have used $\bar{r}(p) \leq 1$. This together with [Lemma 1](#) implies the first assertion of the lemma. The value of the network is then bounded by

$$\begin{aligned} \left| \hat{f}_{\mathcal{P}_2}(\mathbf{x}) \right| &\leq \left| \hat{f}_{\mathcal{P}_2}(\mathbf{x}) - T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) \right| + \left| T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) - f(\mathbf{x}) \right| \\ &\quad + |f(\mathbf{x})| \\ &\leq 2 \cdot \max \left\{ \sup_{\mathbf{x} \in [-a,a]^d} |f(\mathbf{x})|, 1 \right\}, \end{aligned}$$

where we have used that

$$M^{2p} \geq c_{36} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a,a]^d)} \right\} \right)^{4(q+1)}$$

and $M^{2p} \geq c_{32} \cdot (2 \cdot a \cdot d)^{2p} \cdot C$.

In a *last step of the proof* we analyze the bound of $\hat{f}_{\mathcal{P}_2}(\mathbf{x})$ in case that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0.$$

Then the networks $\hat{f}_{test}$ and $\hat{f}_{ind,C_{i,1}}$ ($i \in \{1, \dots, M^d\}$) are not exact (see [Lemma 6](#)). For $\mathbf{x} \in C_{i,1}$ ($i \in \{1, \dots, M^d\}$) this implies

$$\left| \hat{\phi}_{3,1}^{(\mathbf{1},j)} \right| \leq \left| (\partial^{\mathbf{1}} f) \left((\tilde{C}_{j,i})_{left} \right) \right| \quad (j \in \{1, \dots, M^d\})$$

and

$$\left| \hat{\phi}_{2,1}^{(s)} \right| \leq a \quad (s \in \{1, \dots, d\}).$$

Since $\hat{f}_{test}$ produces for at most one summand in [\(20\)](#) a value not equal to zero, this leads to

$$\left| \hat{\phi}_{3,2}^{(\mathbf{1})} \right| \leq \|f\|_{C^q([-a,a]^d)}$$

and

$$\left| \hat{\phi}_{2,2}^{(s)} \right| \leq a, \quad (s \in \{1, \dots, d\}).$$

We conclude

$$\begin{aligned}
\left| \hat{f}_{\mathcal{P}_2}(\mathbf{x}) \right| &\leq \left| \hat{f}_p(\mathbf{z}, y_1, \dots, y_{(d+q)_d}) - p(\mathbf{z}, y_1, \dots, y_{(d+q)_d}) \right| \\
&\quad + \left| p(\mathbf{z}, y_1, \dots, y_{(d+q)_d}) \right| \\
&\leq 1 + \sum_{0 \leq \|\mathbf{l}\|_1 \leq q} \frac{1}{\mathbf{l}!} \cdot \|f\|_{C^q([-a, a]^d)} \cdot (2a)^{\|\mathbf{l}\|_1} \\
&\leq 1 + \|f\|_{C^q([-a, a]^d)} \cdot \left(\sum_{l=0}^{\infty} \frac{(2a)^l}{l!} \right)^d \\
&= 1 + e^{2ad} \cdot \|f\|_{C^q([-a, a]^d)}.
\end{aligned}$$

□

A.1.6. *Key step 3: Approximating of $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ by wide neural networks.*
In order to approximate $f(\mathbf{x})$ in supremum norm, we will use the neural network $\hat{f}_{\mathcal{P}_2}$ of [Lemma 3](#) to construct a network which approximates

$$w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}),$$

where

$$(22) \quad w_{\mathcal{P}_2}(\mathbf{x}) = \prod_{j=1}^d \left(1 - \frac{M^2}{a} \cdot \left| (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} + \frac{a}{M^2} - x^{(j)} \right| \right)_+$$

is a linear tensorproduct B-spline which takes its maximum value at the center of $C_{\mathcal{P}_2}(\mathbf{x})$, which is nonzero in the inner part of $C_{\mathcal{P}_2}(\mathbf{x})$ and which vanishes outside of $C_{\mathcal{P}_2}(\mathbf{x})$. It is easy to see that $w_{\mathcal{P}_2}(\mathbf{x})$ is less than or equal to $1/M^{2p}$ in case that $\mathbf{x}$ is contained in

$$\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0.$$

Since $w_{\mathcal{P}_2}(\mathbf{x})$ is close to zero close to the boundary of $C_{\mathcal{P}_2}(\mathbf{x})$ it will be possible to construct this neural network such that it approximates $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ in supremum norm.

LEMMA 7. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$ and $M \in \mathbb{N}_0$ sufficiently large (independent of the size of a , but*

$$M^{2p} \geq \max\{c_{36}, c_{33}\} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)}$$

and

$$M^{2p} \geq c_{32} \cdot (2 \cdot a \cdot d)^{2p} \cdot C$$

must hold). Let $p = q + s$ for some $q \in \mathbb{N}_0$, $s \in (0, 1]$ and let $C > 0$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function and let $w_{\mathcal{P}_2}$ be defined as in (22). Then there exists a network

$$\hat{f} \in \mathcal{F}(L, r)$$

with

$$L = 5 + \lceil \log_4(M^{2p}) \rceil \cdot (\lceil \log_2(\max\{q, d\} + 1) \rceil + 1)$$

and

$$r = 64 \cdot \binom{d+q}{d} \cdot d^2 \cdot (q+1) \cdot M^d$$

such that

$$\left| \hat{f}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \leq c_{39} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}$$

holds for $\mathbf{x} \in [-a, a]^d$.

To show this result we need again some auxiliary lemmata. [Lemma 9](#) below shows that each weight $w_{\mathcal{P}_2}(\mathbf{x})$ (see (22)) can be approximated by a neural network in case that $\mathbf{x}$ does not lie close to the boundaries of a cube of $\mathcal{P}_2$. The values of $(C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)}$ are computed as described in $\hat{\phi}_{2,2}$ in the proof of [Lemma 3](#). Furthermore we use that

$$\begin{aligned} & \left(1 - \frac{M^2}{a} \cdot \left| (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} + \frac{a}{M^2} - x^{(j)} \right| \right)_+ \\ &= \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} \right) \right)_+ \\ & \quad - 2 \cdot \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} - \frac{a}{M^2} \right) \right)_+ \\ & \quad + \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} - \frac{2 \cdot a}{M^2} \right) \right)_+, \quad j \in \{1, \dots, d\}. \end{aligned}$$

Thus every factor can be easily computed by applying the ReLU activation function. The final product is approximated by using the following lemma.

LEMMA 8. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Then for any $R \in \mathbb{N}$ and any $a \geq 1$ a neural network*

$$\hat{f}_{mult,d} \in \mathcal{F}(R \cdot \lceil \log_2(d) \rceil, 18d)$$

exists such that

$$\left| \hat{f}_{mult,d}(\mathbf{x}) - \prod_{i=1}^d x^{(i)} \right| \leq 4^{4d+1} \cdot a^{4d} \cdot d \cdot 4^{-R}$$

holds for all $\mathbf{x} \in [-a, a]^d$.

PROOF. A similar result can be found in Lemma A.3 in the Supplement of [2]. For sake of completeness we provide a proof of our result in Supplement B. $\square$

LEMMA 9. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$ and $M \geq 4^{4d+1} \cdot d$. Let $\mathcal{P}_2$ be the partition defined in (15) and let $w_{\mathcal{P}_2}(\mathbf{x})$ be the corresponding weight defined by (22). Then there exists a neural network*

$$\hat{f}_{w_{\mathcal{P}_2}} \in \mathcal{F}(5 + \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(d) \rceil, r),$$

with

$$r = \max \left\{ 18d, 2d + d \cdot M^d \cdot 2 \cdot (2 + 2d) \right\}$$

such that

$$\left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \right| \leq 4^{4d+1} \cdot d \cdot \frac{1}{M^{2p}}$$

for $\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0$ and

$$|\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})| \leq 2$$

for $\mathbf{x} \in [-a, a]^d$.

PROOF. The first four hidden layers of $\hat{f}_{w_{\mathcal{P}_2}}$ compute the value of

$$(C_{\mathcal{P}_2}(\mathbf{x}))_{left}$$

and shift the value of $\mathbf{x}$ in the next hidden layer, respectively. This can be done as described in $\hat{\phi}_{1,2}$ and $\hat{\phi}_{2,2}$ in the proof of Lemma 3 with $2d + d \cdot$

$M^d \cdot 2 \cdot (2 + 2d)$ neurons per layer. The fifth hidden layer then computes the functions

$$\begin{aligned} & \left(1 - \frac{M^2}{a} \cdot \left| (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} + \frac{a}{M^2} - x^{(j)} \right| \right)_+ \\ &= \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} \right) \right)_+ \\ & \quad - 2 \cdot \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} - \frac{a}{M^2} \right) \right)_+ \\ & \quad + \left(\frac{M^2}{a} \cdot \left(x^{(j)} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}^{(j)} - \frac{2 \cdot a}{M^2} \right) \right)_+, \quad j \in \{1, \dots, d\}, \end{aligned}$$

using the networks

$$\begin{aligned} \hat{f}_{w_{\mathcal{P}_2,j}}(\mathbf{x}) &= \sigma \left(\frac{M^2}{a} \cdot \left(\hat{\phi}_{1,2}^{(j)} - \hat{\phi}_{2,2}^{(j)} \right) \right) \\ & \quad - 2 \cdot \sigma \left(\frac{M^2}{a} \cdot \left(\hat{\phi}_{1,2}^{(j)} - \hat{\phi}_{2,2}^{(j)} - \frac{a}{M^2} \right) \right) \\ & \quad + \sigma \left(\frac{M^2}{a} \cdot \left(\hat{\phi}_{1,2}^{(j)} - \hat{\phi}_{2,2}^{(j)} - \frac{2 \cdot a}{M^2} \right) \right) \end{aligned}$$

with $3d$ neurons. The product of $w_{\mathcal{P}_2,j}(\mathbf{x})$ ($j \in \{1, \dots, d\}$) can then be computed by the network $\hat{f}_{mult,d}$ of [Lemma 8](#), where we choose $x^{(j)} = \hat{f}_{w_{\mathcal{P}_2,j}}(\mathbf{x})$. Finally we set

$$\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) = \hat{f}_{mult,d} \left(\hat{f}_{w_{\mathcal{P}_2,1}}(\mathbf{x}), \dots, \hat{f}_{w_{\mathcal{P}_2,d}}(\mathbf{x}) \right).$$

By choosing $R = \lceil \log_4(M^{2p}) \rceil$ in [Lemma 8](#), this network lies in the class

$$\mathcal{F} \left(4 + 1 + \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(d) \rceil, \max \left\{ 18d, 2d + d \cdot M^d \cdot 2 \cdot (2 + 2d), 3d \right\} \right),$$

and according to [Lemma 8](#) (where we set $a = 1$) it approximates $w_{\mathcal{P}_2}(\mathbf{x})$ with an error of size

$$4^{4d+1} \cdot d \cdot \frac{1}{M^{2p}}$$

in case that $\mathbf{x}$ is contained in $\bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0$. Since $|\hat{f}_{w_{\mathcal{P}_2,j}}(\mathbf{x})| \leq 1$ for $j \in \{1, \dots, d\}$ we can bound the value of the network using triangle inequality by

$$|\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})| \leq \left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) - \prod_{j=1}^d \hat{f}_{w_{\mathcal{P}_2,j}}(\mathbf{x}) \right| + \left| \prod_{j=1}^d \hat{f}_{w_{\mathcal{P}_2,j}}(\mathbf{x}) \right| \leq 2$$

for $\mathbf{x} \in [-a, a)^d$, where we have used that

$$M^{2p} \geq 4^{4d+1} \cdot d.$$

□

The networks $\hat{f}_{\mathcal{P}_2}$ of [Lemma 3](#) and $\hat{f}_{w_{\mathcal{P}_2}}$ of [Lemma 9](#) are only good approximation of $f(\mathbf{x})$ and $w_{\mathcal{P}_2}(\mathbf{x})$, respectively, in case that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

The following network helps us to control the "large" approximation error in case that

$$(23) \quad \mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0.$$

For instance, this network will be 1 in case that $\mathbf{x}$ is contained in (23) and 0 if $\mathbf{x}$ is contained in

$$\bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{2/M^{2p+2}}^0.$$

Thus we say that this network *checks* the position of our input $\mathbf{x}$.

A straightforward way to approximate

$$\mathbb{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x}) = 1 - \sum_{k \in \{1, \dots, M^{2d}\}} \mathbb{1}_{(C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

by a neural network would be to approximate each of the M^{2d} indicator functions by networks, respectively. In this case the overall number of neurons per layer would be of order M^{2d} . That is why we again use a two scale approximation. In a first step we compute the position of $(C_{\mathcal{P}_1}(\mathbf{x}))_{left}$ as described in $\hat{\phi}_{2,1}$ in [Lemma 3](#). Let $i \in \{1, \dots, M^d\}$ such that $(C_{\mathcal{P}_1}(\mathbf{x})) = C_{i,1}$. In a second step we then only need to approximate

$$\mathbb{1}_{\bigcup_{j \in \{1, \dots, M^d\}} \tilde{C}_{j,i} \setminus (\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x}) = 1 - \sum_{j \in \{1, \dots, M^d\}} \mathbb{1}_{(\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x})$$

which can be done by $c_{40} \cdot M^d$ neurons per layer.

Since $\hat{\phi}_{2,1}$ is only a good approximation in case that

$$\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^d\}} C_{k,1} \setminus (C_{k,1})_{1/M^{2p+2}}^0$$

we further need to check whether $\mathbf{x}$ is close to the boundaries of the coarse grid of $\mathcal{P}_1$, i.e.

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^d\}} C_{k,1} \setminus (C_{k,1})_{1/M^{2p+2}}^0.$$

This can be done by computing

$$(24) \quad \mathbb{1}_{\bigcup_{k \in \{1, \dots, M^d\}} C_{k,1} \setminus (C_{k,1})_{1/M^{2p+2}}^0} = 1 - \sum_{k \in \{1, \dots, M^d\}} \mathbb{1}_{(C_{k,1})_{1/M^{2p+2}}^0}(\mathbf{x}),$$

with the networks of [Lemma 6 a](#)). Combining this, our final network then computes

$$(25) \quad 1 - \sigma(1 - \mathbb{1}_{\bigcup_{j \in \{1, \dots, M^d\}} \tilde{C}_{j,i}/(\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x}) - \mathbb{1}_{\bigcup_{k \in \{1, \dots, M^d\}} C_{k,1}/(C_{k,1})_{1/M^{2p+2}}^0}(\mathbf{x}))$$

where we will exploit the fact of ReLU activation function, that it is zero in case of negativ input. In particular the second indicator function in (25) is computed by using the networks $\hat{f}_{ind}$ of [Lemma 6 a](#)), while the first indicator function is approximated by using the networks $\hat{f}_{test}$ of [Lemma 6 b](#)).

LEMMA 10. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$. Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be the partitions defined in (15) and let $M \in \mathbb{N}$. Then there exists a neural network*

$$\hat{f}_{check, \mathcal{P}_2} \in \mathcal{F}\left(5, 2d + (4d^2 + 4d) \cdot M^d\right)$$

satisfying

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = \mathbb{1}_{\bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

for $\mathbf{x} \notin \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0 \setminus (C_{i,2})_{2/M^{2p+2}}^0$ and

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) \in [0, 1]$$

for $\mathbf{x} \in [-a, a)^d$.

PROOF. Throughout the proof we assume that $i \in \{1, \dots, M^d\}$ satisfies $C_{\mathcal{P}_1}(\mathbf{x}) = C_{i,1}$. The above described two scale approximation proceeds as follows: In the first part of the network we check whether $\mathbf{x}$ is contained in

$$\bigcup_{k \in \{1, \dots, M^d\}} C_{k,1} \setminus (C_{k,1})_{1/M^{2p+2}}^0.$$

Therefore our network approximates in the first two hidden layers the function

$$f_1(\mathbf{x}) = \mathbb{1}_{\bigcup_{i \in \{1, \dots, M^d\}} C_{i,1} \setminus (C_{i,1})_{1/M^{2p+2}}^0}(\mathbf{x}) = 1 - \sum_{i \in \{1, \dots, M^d\}} \mathbb{1}_{(C_{i,1})_{1/M^{2p+2}}^0}(\mathbf{x})$$

by

$$\hat{f}_1(\mathbf{x}) = 1 - \sum_{k \in \{1, \dots, M^d\}} \hat{f}_{ind, (C_{k,1})_{1/M^{2p+2}}^0}(\mathbf{x}),$$

where $\hat{f}_{ind, (C_{k,1})_{1/M^{2p+2}}^0}$ are the networks of [Lemma 6](#), which need $2d$ neurons per layer, respectively. To approximate the indicator functions on the partition $\mathcal{P}_2$ only for the cubes $C_{k,2} \subset C_{\mathcal{P}_1}(\mathbf{x})$, we further need to compute the position of $(C_{\mathcal{P}_1}(\mathbf{x}))_{left}$. This can be done as described by the network $\hat{\phi}_{2,1}$ in the proof of [Lemma 3](#) with $d \cdot M^d \cdot 2d$ neurons. To shift the value of $\mathbf{x}$ in the next hidden layers we further apply the network $\hat{f}_{id}^2$, which needs $2d$ neurons per layer. Analogous to (17) we can describe the cubes $(\tilde{C}_{j,i})_{1/M^{2p+2}}^0$ ($j \in \{1, \dots, M^d\}$) that are contained in the cube $C_{i,1}$, by

$$\begin{aligned} (\mathcal{A}^{(j)})_{1/M^{2p+2}}^0 &= \left\{ \mathbf{x} \in \mathbb{R}^d : -x^{(k)} + \phi_{2,1}^{(k)} + v_j^{(k)} + \frac{1}{M^{2p+2}} \leq 0 \right. \\ &\quad \left. \text{and } x^{(k)} - \phi_{2,1}^{(k)} - v_j^{(k)} - \frac{2a}{M^2} + \frac{1}{M^{2p+2}} < 0 \text{ for all } k \in \{1, \dots, d\} \right\}. \end{aligned}$$

Then the function

$$f_2(\mathbf{x}) = \mathbb{1}_{\bigcup_{j \in \{1, \dots, M^d\}} \tilde{C}_{j,i} \setminus (\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x}) = 1 - \sum_{j \in \{1, \dots, M^d\}} \mathbb{1}_{(\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x})$$

can be approximated by

$$\begin{aligned} \hat{f}_2(\mathbf{x}) &= 1 - \sum_{j \in \{1, \dots, M^d\}} \hat{f}_{test} \left(\hat{f}_{id}^2(\mathbf{x}), \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{1}{M^{2p+2}} \cdot \mathbf{1}, \right. \\ &\quad \left. \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1} - \frac{1}{M^{2p+2}} \cdot \mathbf{1}, 1 \right), \end{aligned}$$

where $\hat{f}_{test}$ is the network of [Lemma 6](#) b), which needs 2 hidden layers and $2 \cdot (2d+2)$ neurons per layer. Combining the networks $\hat{f}_1$ and $\hat{f}_2$ and using the characteristics of ReLU activation function, that is zero in case of negative input, finally let us approximate

$$\mathbb{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

by

$$\begin{aligned}
\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) &= 1 - \sigma \left(1 - \hat{f}_2(\mathbf{x}) - \hat{f}_{id}^2 \left(\hat{f}_1(\mathbf{x}) \right) \right) \\
&= 1 - \sigma \left(\sum_{j \in \{1, \dots, M^d\}} \hat{f}_{test} \left(\hat{f}_{id}^2(\mathbf{x}), \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{1}{M^{2p+2}} \cdot \mathbf{1}, \right. \right. \\
&\quad \left. \left. \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1} - \frac{1}{M^{2p+2}} \cdot \mathbf{1}, 1 \right) \right. \\
&\quad \left. - \hat{f}_{id}^2 \left(1 - \sum_{k \in \{1, \dots, M^d\}} \hat{f}_{ind, (C_{k,1})_{1/M^{2p+2}}^0}(\mathbf{x}) \right) \right).
\end{aligned}$$

Now it is easy to see, that our whole network is contained in the network class

$$\mathcal{F}(5, r)$$

with

$$\begin{aligned}
r &= \max\{2d + d \cdot M^d \cdot 2d + M^d \cdot 2d, M^d \cdot 2 \cdot (2 + 2d) + 2\} \\
&\leq 2d + (4d^2 + 4d) \cdot M^d.
\end{aligned}$$

In the following we show that we have

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = \mathbb{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

for $\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0 \setminus (C_{k,2})_{2/M^{2p+2}}^0$. Here we distinguish between *three* cases. In our first case we assume that

$$\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^d\}} (C_{k,1})_{1/M^{2p+2}}^0,$$

which also implies that

$$\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

Furthermore we get from [Lemma 6](#) that $\hat{f}_1(\mathbf{x}) = 1$ from which we can conclude

$$1 - \hat{f}_2(\mathbf{x}) - \hat{f}_{id}^2 \left(\hat{f}_1(\mathbf{x}) \right)$$

$$\begin{aligned}
&= \sum_{j \in \{1, \dots, M^d\}} \hat{f}_{test} \left(\hat{f}_{id}^2(\mathbf{x}), \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{1}{M^{2p+2}} \cdot \mathbf{1}, \right. \\
&\quad \left. \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1} - \frac{1}{M^{2p+2}} \cdot \mathbf{1}, 1 \right) - 1 \\
&\leq 0.
\end{aligned}$$

Here we have used that each $\hat{f}_{test}$ is contained in $[0, 1]$ and that at most one $\hat{f}_{test}$ in the sum is larger than 0. Finally we get

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = 1 - 0 = 1 = \mathbf{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x}).$$

In our second case we assume that

$$(26) \quad \mathbf{x} \in \bigcup_{k \in \{1, \dots, M^d\}} (C_{k,1})_{1/M^{2p+2}}^0.$$

and

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{2/M^{2p+2}}^0.$$

Then we have $\hat{\phi}_{2,1} = (C_{\mathcal{P}_1}(\mathbf{x}))_{left} = (C_{i,1})_{left}$. Furthermore we can conclude that

$$\begin{aligned}
(\mathcal{A}^{(j)})_{1/M^{2p+2}}^0 &= \left\{ \mathbf{x} \in \mathbb{R}^d : -\hat{\phi}_{1,1}^{(k)} + \hat{\phi}_{2,1}^{(k)} + v_j^{(k)} + \frac{1}{M^{2p+2}} \leq 0 \right. \\
&\quad \text{and } \hat{\phi}_{1,1}^{(k)} - \hat{\phi}_{2,1}^{(k)} - v_j^{(k)} - \frac{2a}{M^2} + \frac{1}{M^{2p+2}} < 0 \\
&\quad \left. \text{for all } k \in \{1, \dots, d\} \right\} \\
&= (\tilde{C}_{j,i})_{1/M^{2p+2}}^0
\end{aligned}$$

for $j \in \{1, \dots, M^d\}$. Since we only have to show our assumption for

$$\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0 \setminus (C_{k,2})_{2/M^{2p+2}}^0,$$

we can conclude by [Lemma 6](#) that

$$\hat{f}_{test} \left(\hat{\phi}_{1,1}, \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{1}{M^{2p+2}} \cdot \mathbf{1}, \right.$$

$$\begin{aligned} & \hat{\phi}_{2,1} + \mathbf{v}_j + \frac{2a}{M^2} \cdot \mathbf{1} - \frac{1}{M^{2p+2}} \cdot \mathbf{1}, 1) \\ &= \mathbb{1}_{(\bar{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x}) \end{aligned}$$

for all $j \in \{1, \dots, M^d\}$. This implies

$$\hat{f}_2(\mathbf{x}) = f_2(\mathbf{x}).$$

Since

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{2/M^{2p+2}}^0$$

we can further conclude that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^d\}} (C_{k,1})_{2/M^{2p+2}}^0$$

and it follows by [Lemma 6](#) that

$$\hat{f}_1(\mathbf{x}) = f_1(\mathbf{x}) = 0.$$

Thus we have

$$1 - \hat{f}_2(\mathbf{x}) - \hat{f}_{id}^2(\hat{f}_1(\mathbf{x})) = 1 - f_2(\mathbf{x}) = 1 - 0 = 1$$

and

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = 1 - 1 = 0 = \mathbb{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x}).$$

In our third case we assume [\(26\)](#), but

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2}) \setminus (C_{k,2})_{1/M^{2p+2}}^0,$$

which means that

$$\mathbf{x} \notin \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

In this case the approximation $\hat{f}_1(\mathbf{x})$ is not exact. By [Lemma 6](#) all values of $\hat{f}_{ind, (C_{k,1})_{1/M^{2p+2}}^0}^0$ ($k \in \{1, \dots, M^d\}$) in the definition of $\hat{f}_1$ are contained in $[0, 1]$. Thus we have

$$\hat{f}_1(\mathbf{x}) \in [0, 1].$$

Since (26) holds we further have

$$\hat{f}_2(\mathbf{x}) = f_2(\mathbf{x})$$

as shown in the second case. Summarizing this we can conclude that

$$\begin{aligned} 1 - f_2(\mathbf{x}) - \hat{f}_{id}^2(\hat{f}_1(\mathbf{x})) &= \sum_{j \in \{1, \dots, M^d\}} \mathbb{1}_{(\tilde{C}_{j,i})_{1/M^{2p+2}}^0}(\mathbf{x}) - \hat{f}_{id}^2(\hat{f}_1(\mathbf{x})) \\ &\leq 0 - 0 = 0. \end{aligned}$$

This implies

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = 1 - 0 = 1 = \mathbb{1}_{\bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0}(\mathbf{x}).$$

By construction of the network

$$\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) \in [0, 1]$$

holds for $\mathbf{x} \in [-a, a]^d$. $\square$

Combining the networks of Lemma 3, Lemma 9 and Lemma 10 finally leads to the network which approximates $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ in supremum norm. The main idea is, that we can define a network

$$\begin{aligned} \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) &= \sigma \left(\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - B_{true} \cdot \hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) \right) \\ &\quad - \sigma \left(-\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - B_{true} \cdot \hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) \right), \end{aligned}$$

which will be equal to $f(\mathbf{x})$ as long as $\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{2/M^{2p+2}}^0$ and which will be 0 for $\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0$. Here we use Lemma 10 and the bound of $\hat{f}_{\mathcal{P}_2}(\mathbf{x})$ given in Lemma 3 as value of B_{true} in the following. Again we exploit the properties of ReLU activation function, that is zero in case of negative input. In particular, in case that $\mathbf{x}$ is close to the boundaries, the network $\hat{f}_{check, \mathcal{P}_2}$ is 1, thus $\hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) = 0$. Otherwise $\hat{f}_{check, \mathcal{P}_2}(\mathbf{x})$ is zero and

$$\hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) = \hat{f}_{\mathcal{P}_2}(\mathbf{x}).$$

Finally we multiply this network by the network $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})$ of Lemma 9.

PROOF OF LEMMA 7. Let $\hat{f}_{\mathcal{P}_2}$ be the network of Lemma 3 and let $\hat{f}_{check, \mathcal{P}_2}$ be the network of Lemma 10. By successively applying $\hat{f}_{id}$ to the output of

one of these networks, we can achieve that both networks have the same number of hidden layers, i.e.

$$L = 4 + \max \left\{ \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil, 1 \right\}.$$

We set

$$\begin{aligned} \hat{f}_{\mathcal{P}_2, \text{true}}(\mathbf{x}) &= \sigma \left(\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - B_{\text{true}} \cdot \hat{f}_{\text{check}, \mathcal{P}_2}(\mathbf{x}) \right) \\ &\quad - \sigma \left(-\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - B_{\text{true}} \cdot \hat{f}_{\text{check}, \mathcal{P}_2}(\mathbf{x}) \right), \end{aligned}$$

where

$$B_{\text{true}} = 2 \cdot e^{2ad} \cdot \max \left\{ \|f\|_{C^q([-a, a]^d)}, 1 \right\}.$$

This network is contained in the network class $\mathcal{F}(L, r)$ with

$$L = 5 + \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil$$

and

$$\begin{aligned} r = \max \left\{ \left(\binom{d+q}{d} + d \right) \cdot M^d \cdot 2 \cdot (2+2d) + 2d, 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} \\ + 2d + (4d^2 + 4d) \cdot M^d. \end{aligned}$$

Due to the fact, that the value of $\hat{f}_{\mathcal{P}_2}$ is bounded by B_{true} according to [Lemma 3](#) and that $\hat{f}_{\text{check}, \mathcal{P}_2}(\mathbf{x})$ is 1 in case that $\mathbf{x}$ lies in

$$(27) \quad \bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})_{1/M^{2p+2}}^0,$$

the properties of ReLU activation function imply that the value of $\hat{f}_{\mathcal{P}_2, \text{true}}(\mathbf{x})$ is zero in case that $\mathbf{x}$ is contained in (27). Let $\hat{f}_{w_{\mathcal{P}_2}}$ be the network of [Lemma 9](#). To multiply the network $\hat{f}_{\mathcal{P}_2, \text{true}}$ by $\hat{f}_{w_{\mathcal{P}_2}}$ we use the network

$$\hat{f}_{\text{mult}} \in \mathcal{F}(\lceil \log_4(M^{2p}) \rceil, 18)$$

of [Lemma 4](#), which satisfies

$$(28) \quad \left| \hat{f}_{\text{mult}}(x, y) - xy \right| \leq 8 \cdot \left(\max \left\{ \|f\|_{\infty, [-a, a]^d}, 1 \right\} \right)^2 \cdot \frac{1}{M^{2p}}$$

for all x, y contained in

$$\left[-2 \cdot \max \left\{ \|f\|_{\infty, [-a, a]^d}, 1 \right\}, 2 \cdot \max \left\{ \|f\|_{\infty, [-a, a]^d}, 1 \right\} \right].$$

Here we have chosen $R = \lceil \log_4(M^{2p}) \rceil$ in [Lemma 4](#).

By successively applying $\hat{f}_{id}$ to the outputs of the networks $\hat{f}_{w_{\mathcal{P}_2}}$ and $\hat{f}_{\mathcal{P}_2, true}$, we can synchronize their depth such that both networks have

$$L = 5 + \lceil \log_4(M^{2p}) \rceil \cdot (\lceil \log_2(\max\{q, d\} + 1) \rceil)$$

hidden layers.

The final network is given by

$$\hat{f}(\mathbf{x}) = \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) \quad (v \in \{1, \dots, 2^d\}).$$

and the network is contained in the network class $\mathcal{F}(L, r)$ with

$$L = 5 + \lceil \log_4(M^{2p}) \rceil \cdot (\lceil \log_2(\max\{q, d\} + 1) \rceil + 1)$$

and

$$\begin{aligned} r &= \max \left\{ \left(\binom{d+q}{d} + d \right) \cdot M^d \cdot 2 \cdot (2+2d) + 2d, 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} \\ &\quad + 8M^d + 2 + \max \left\{ 18d, 2d + d \cdot M^d \cdot 2 \cdot (2+2d) \right\} \\ &\leq 64 \cdot \binom{d+q}{d} \cdot d^2 \cdot (q+1) \cdot M^d. \end{aligned}$$

In case that

$$\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{2/M^{2p+2}}^0,$$

the value of $\mathbf{x}$ is neither contained in

$$(29) \quad \bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})_{1/M^{2p+2}}^0$$

nor contained in

$$(30) \quad \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0 \setminus (C_{i,2})_{2/M^{2p+2}}^0.$$

Thus the network $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})$ approximates $w_{\mathcal{P}_2}(\mathbf{x})$ according to [Lemma 9](#) with an error of size

$$(31) \quad 4^{d+1} \cdot d \cdot \frac{1}{M^{2p}}$$

and $\hat{f}_{\mathcal{P}_2}(\mathbf{x})$ approximates $f(\mathbf{x})$ according to [Lemma 3](#) with an error of size

$$(32) \quad c_{33} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a,a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}.$$

Since $\hat{f}_{check, \mathcal{P}_2}(\mathbf{x}) = 0$, we have

$$\hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) = \sigma(\hat{f}_{\mathcal{P}_2}(\mathbf{x})) - \sigma(-\hat{f}_{\mathcal{P}_2}(\mathbf{x})) = \hat{f}_{\mathcal{P}_2}(\mathbf{x}).$$

Since $M^{2p} \geq 4^{d+1} \cdot d$, we can bound the value of $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})$ using triangle inequality by

$$|\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})| \leq |\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x})| + |w_{\mathcal{P}_2}(\mathbf{x})| \leq 2.$$

Furthermore we can bound

$$|\hat{f}_{\mathcal{P}_2}(\mathbf{x})| \leq |\hat{f}_{\mathcal{P}_2}(\mathbf{x}) - f(\mathbf{x})| + |f(\mathbf{x})| \leq 2 \cdot \|f\|_{\infty, [-a,a]^d},$$

where we used $M^{2p} \geq c_{33} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a,a]^d)} \right\} \right)^{4(q+1)}$. Thus both networks are contained in the interval, where [\(28\)](#) holds. Using triangle inequality, this implies

$$\begin{aligned} & \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\ & \leq \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2}(\mathbf{x}) \right| \\ & \quad + \left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2}(\mathbf{x}) \right| \\ & \quad + \left| w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\ & \leq c_{41} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a,a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}. \end{aligned}$$

In case that $\mathbf{x}$ is contained in [\(29\)](#) the approximation error of $\hat{f}_{\mathcal{P}_2}$ is not of size $1/M^{2p}$. But the value of $\hat{f}_{check, \mathcal{P}_2}(\mathbf{x})$ is 1, such that $\hat{f}_{\mathcal{P}_2, true}$ is zero. Furthermore we have

$$\left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \right| \leq 2.$$

Thus $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})$ and $\hat{f}_{\mathcal{P}_2, true}(\mathbf{x})$ are contained in the interval, where [\(28\)](#) holds. Together with

$$0 \leq w_{\mathcal{P}_2}(\mathbf{x}) \leq \frac{1}{a \cdot M^{2p}}$$

and the triangle inequality it follows

$$\begin{aligned}
& \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\
& \leq \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \\
& \quad + \left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \\
& \quad + \left| w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\
& \leq c_{42} \cdot \left(\max\{\|f\|_{\infty, [-a, a]^d}, 1\} \right)^2 \cdot \frac{1}{M^{2p}}.
\end{aligned}$$

In case that $\mathbf{x}$ is in (30) but not in (29) the network $\hat{f}_{\mathcal{P}_2}$ approximates $f(\mathbf{x})$ with an error as in (32). Furthermore, $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \in [-2, 2]$ approximates $w_{\mathcal{P}_2}(\mathbf{x})$ with an error as in (31). The value of $\hat{f}_{check, \mathcal{P}_2}(\mathbf{x})$ is contained in the interval $[0, 1]$, such that

$$\left| \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \leq \left| \hat{f}_{\mathcal{P}_2}(\mathbf{x}) \right| \leq 2 \cdot \max\left\{ \|f\|_{\infty, [-a, a]^d}, 1 \right\}.$$

Hence $\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x})$ and $\hat{f}_{\mathcal{P}_2, true}(\mathbf{x})$ are contained in the interval, where (28) holds. Together with

$$w_{\mathcal{P}_2}(\mathbf{x}) \leq \frac{2}{a \cdot M^{2p}}$$

and the triangle inequality it follows again

$$\begin{aligned}
& \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\
& \leq \left| \hat{f}_{mult} \left(\hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}), \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right) - \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \\
& \quad + \left| \hat{f}_{w_{\mathcal{P}_2}}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \\
& \quad + \left| w_{\mathcal{P}_2}(\mathbf{x}) \cdot \hat{f}_{\mathcal{P}_2, true}(\mathbf{x}) \right| \\
& \leq c_{43} \cdot \left(\max\{\|f\|_{\infty, [-a, a]^d}, 1\} \right)^2 \cdot \frac{1}{M^{2p}}.
\end{aligned}$$

□

A.1.7. *Key step 4: Applying $\hat{f}$ to slightly shifted partitions.* Finally we will use a finite sum of those networks of Lemma 7 constructed to 2^d slightly shifted versions of $\mathcal{P}_2$ in order to approximate $f(\mathbf{x})$. This shows Theorem 2 a).

PROOF OF THEOREM 2 A). By increasing a , if necessary, it suffices to show that there exists a network $\hat{f}_{wide}$ satisfying

$$\begin{aligned} & \sup_{x \in [-a/2, a/2]^d} \left| f(\mathbf{x}) - \hat{f}_{wide}(\mathbf{x}) \right| \\ & \leq c_{44} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}. \end{aligned}$$

Let $\mathcal{P}_1$ and $\mathcal{P}_2$ be the partitions defined as in (15). We set

$$\mathcal{P}_{1,1} = \mathcal{P}_1 \text{ and } \mathcal{P}_{2,1} = \mathcal{P}_2$$

and define for each $v \in \{2, \dots, 2^d\}$ partitions $\mathcal{P}_{1,v}$ and $\mathcal{P}_{2,v}$, which are modifications of $\mathcal{P}_{1,1}$ and $\mathcal{P}_{2,1}$ where at least one of the components is shifted by a/M^2 . The idea is illustrated for the case $d = 2$ in Fig. 2. Here one sees,

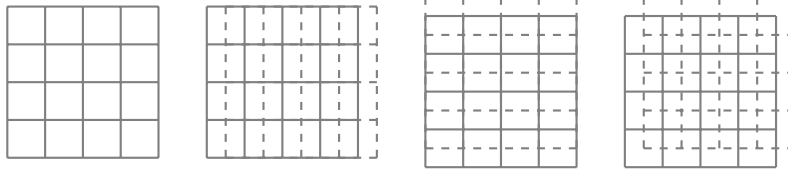


FIG 2. 2^2 different partitions in the case $d = 2$

that for $d = 2$ there exist $2^2 = 4$ different partitions, if we shift our partition along at least one component by the same additional distance. We denote by $C_{k,2,v}$ the corresponding cubes of the partition $\mathcal{P}_{2,v}$ ($k \in \{1, \dots, M^{2d}\}$).

The idea of the proof of Theorem 2 a) is to compute a linear combination of networks $\hat{f}_{\mathcal{P}_{2,1}}, \dots, \hat{f}_{\mathcal{P}_{2,2^d}}$ of Lemma 3 (where the $\mathcal{P}_{2,v}$ are treated as $\mathcal{P}_2$ in Lemma 3, respectively). To avoid that the approximation error of the networks increases close to the boundaries of some cube of the partitions, we multiply each value of $\hat{f}_{\mathcal{P}_{2,v}}$ with a weight

$$(33) \quad w_v(\mathbf{x}) = \prod_{j=1}^d \left(1 - \frac{M^2}{a} \cdot \left| (C_{\mathcal{P}_{2,v}}(\mathbf{x}))_{left}^{(j)} + \frac{a}{M^2} - x^{(j)} \right| \right)_+.$$

It is easy to see that $w_v(\mathbf{x})$ is a linear tensorproduct B-spline which takes its maximum value at the center of $C_{\mathcal{P}_{2,v}}(\mathbf{x})$, which is nonzero in the inner part of $C_{\mathcal{P}_{2,v}}(\mathbf{x})$ and which vanishes outside of $C_{\mathcal{P}_{2,v}}(\mathbf{x})$. Consequently we have $w_1(\mathbf{x}) + \dots + w_{2^d}(\mathbf{x}) = 1$ for $x \in [-a/2, a/2]^d$ (since $a/M^2 \leq a/2$). Let $\hat{f}_1, \dots, \hat{f}_{2^d}$ be the networks of Lemma 7 corresponding to the partitions $\mathcal{P}_{1,v}$

and $\mathcal{P}_{2,v}$ ($v \in \{1, \dots, 2^d\}$), respectively. Since $[-a/2, a/2]^d \subset [-a+a/M^2, a]^d$ each $\mathcal{P}_{1,v}$ and $\mathcal{P}_{2,v}$ form a partition of a set which contains $[-a/2, a/2]^d$ and the error bounds of [Lemma 7](#) hold for each network $\hat{f}_v$ on $[-a/2, a/2]^d$. We set

$$\hat{f}_{wide}(\mathbf{x}) = \sum_{v=1}^{2^d} \hat{f}_v(\mathbf{x}).$$

Using [Lemma 7](#) it is easy to see that this network is contained in the network class $\mathcal{F}(L, r)$ with

$$L = 5 + \lceil \log_4(M^{2p}) \rceil \cdot (\lceil \log_2(\max\{q, d\} + 1) \rceil + 1)$$

and

$$r = 2^d \cdot 64 \cdot \binom{d+q}{d} \cdot d^2 \cdot (q+1) \cdot M^d.$$

Since

$$f(\mathbf{x}) = \sum_{v=1}^{2^d} w_v(\mathbf{x}) \cdot f(\mathbf{x})$$

it follows directly by [Lemma 7](#)

$$\begin{aligned} & \left| \hat{f}_{wide}(\mathbf{x}) - f(\mathbf{x}) \right| \\ &= \left| \sum_{v=1}^{2^d} \hat{f}_{mult} \left(\hat{f}_{w_v}(\mathbf{x}), \hat{f}_{\mathcal{P}_{2,v}, true}(\mathbf{x}) \right) - \sum_{v=1}^{2^d} w_v(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\ &\leq \sum_{v=1}^{2^d} \left| \hat{f}_{mult} \left(\hat{f}_{w_v}(\mathbf{x}), \hat{f}_{\mathcal{P}_{2,v}, true}(\mathbf{x}) \right) - w_v(\mathbf{x}) \cdot f(\mathbf{x}) \right| \\ &\leq c_{45} \cdot \left(\max \left\{ 2a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}. \end{aligned}$$

□

A.1.8. *Idea of the proof of Theorem 2 b).* As in the proof of Theorem 2 a) the approximation of a piecewise Taylor polynomial is essential. This follows again by [Lemma 1](#) since

$$\sup_{\mathbf{x} \in [-a, a]^d} \left| f(\mathbf{x}) - T_{f, q, ((C_{\mathcal{P}_2}(\mathbf{x}))_{left})}(\mathbf{x}) \right| \leq c_{32} \cdot C \cdot (2 \cdot a \cdot d)^p \cdot \frac{1}{M^{2p}}.$$

Again we denote by $\tilde{C}_{1,i}, \dots, \tilde{C}_{M^d,i}$ those cubes of $\mathcal{P}_2$ that are contained in $C_{i,1}$. But this time we order the cubes in such a way that we have $(\tilde{C}_{1,i})_{left} = (C_{i,1})_{left}$ and that

$$(34) \quad (\tilde{C}_{k,i})_{left} = (\tilde{C}_{k-1,i})_{left} + \tilde{\mathbf{v}}_k$$

holds for all $k \in \{2, \dots, M^d\}, i \in \{1, \dots, M^d\}$ and some vector $\tilde{\mathbf{v}}_k$ with entries in $\{0, 2a/M^2\}$ where exactly one entry is different to zero. Here the vector $\tilde{\mathbf{v}}_k$ describes the position of $(C_{k,i})_{left}$ relative to $(C_{k-1,i})_{left}$ and again we order the cubes in such a way that the position is independent of i .

To compute $T_{f,q,((C_{\mathcal{P}_2}(\mathbf{x}))_{left})}(\mathbf{x})$ the very deep neural network of Theorem 2 b) proceeds in two steps: In a first step it computes $(C_{\mathcal{P}_1}(\mathbf{x}))_{left}$ and the values of

$$(\partial^{\mathbf{l}} f)((C_{i,1})_{left})$$

for each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and suitably defined numbers

$$b_{k,i}^{(1)} \in \mathbb{Z}, \quad |b_{k,i}^{(1)}| \leq e^d + 1 \quad (k \in \{1, \dots, M^d\}),$$

which depend on $C_{i,1}$ for $i \in \{1, \dots, M^d\}$. Assume that $\mathbf{x} \in C_{i,1}$ for some $i \in \{1, \dots, M^d\}$. In the second step the neural network successively computes approximations

$$(\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{k,i})_{left}), \quad k \in \{1, \dots, M^d\}$$

of

$$(\partial^{\mathbf{l}} f)((\tilde{C}_{k,i})_{left})$$

for each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. To do this we start with

$$(\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{1,i})_{left}) = (\partial^{\mathbf{l}} f)((C_{\mathcal{P}_1}(\mathbf{x}))_{left}).$$

By construction of the first step and since $(\tilde{C}_{1,i})_{left} = (C_{\mathcal{P}_1}(\mathbf{x}))_{left}$ these estimates have error zero. As soon as we have computed the above estimates for some $k \in \{1, \dots, M^d - 1\}$ we use the Taylor polynomials with these coefficients around $(\tilde{C}_{k,i})_{left}$ in order to compute

$$\sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} \hat{f})((\tilde{C}_{k,i})_{left})}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}}$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and we define

$$\begin{aligned} (\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{k+1,i})_{left}) &= \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} \hat{f})((\tilde{C}_{k,i})_{left})}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}} \\ &\quad + b_{k,i}^{(1)} \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1} \end{aligned}$$

where

$$c_{46} = C \cdot d^p \cdot \max\{c_{32}(q, d), c_{32}(q-1, d), \dots, c_{32}(0, d)\}$$

(and c_{32} is the constant of Lemma 1). Assume that

$$\left| (\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{k,i})_{left}) - (\partial^{\mathbf{l}} f)((\tilde{C}_{k,i})_{left}) \right| \leq c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1}$$

holds for all $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ (which holds by construction for $k = 1$). Then

$$\begin{aligned} &\left| \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} \hat{f})((\tilde{C}_{k,i})_{left})}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}} \right. \\ &\quad \left. - (\partial^{\mathbf{l}} f)((\tilde{C}_{k+1,i})_{left}) \right| \\ &\leq \left| \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} \hat{f})((\tilde{C}_{k,i})_{left})}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}} \right. \\ &\quad \left. - \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} f)((\tilde{C}_{k,i})_{left})}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}} \right| \\ &\quad + \left| \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{(\partial^{\mathbf{l}+\mathbf{j}} f)(\tilde{C}_{k,i})_{left}}{\mathbf{j}!} \cdot \left((\tilde{C}_{k+1,i})_{left} - (\tilde{C}_{k,i})_{left} \right)^{\mathbf{j}} \right. \\ &\quad \left. - (\partial^{\mathbf{l}} f)((\tilde{C}_{k+1,i})_{left}) \right| \\ &\leq \sum_{\substack{\mathbf{j} \in \mathbb{N}_0^d: \\ \|\mathbf{j}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{1}{\mathbf{j}!} \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}+\mathbf{j}\|_1} \cdot \left(\frac{2a}{M^2} \right)^{\|\mathbf{j}\|_1} + c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1} \end{aligned}$$

$$\leq (c_{46} \cdot e^d + c_{46}) \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1}.$$

This implies that we can choose $b_{k,i}^{(\mathbf{l})} \in \mathbb{Z}$ such that

$$|b_{k,i}^{(\mathbf{l})}| \leq e^d + 1$$

and

$$\left| (\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{k+1,i})_{left}) - (\partial^{\mathbf{l}} f)((\tilde{C}_{k+1,i})_{left}) \right| \leq c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1}.$$

Observe that in this way we have defined the coefficients $b_{k,i}^{(\mathbf{l})}$ for each cube $C_{i,1}$. We will encode these coefficients for each $i \in \{1, \dots, M^d\}$ and each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ in the single number

$$b_i^{(\mathbf{l})} = \sum_{k=1}^{M^d-1} \left(b_{k,i}^{(\mathbf{l})} + \lceil e^d \rceil + 2 \right) \cdot (4 + 2\lceil e^d \rceil)^{-k} \in [0, 1].$$

In a last step the neural network then computes

$$(35) \quad \hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) := \sum_{\substack{\mathbf{l} \in \mathbb{N}_0^d: \\ \|\mathbf{l}\|_1 \leq q}} \frac{(\partial^{\mathbf{l}} \hat{f})((C_{\mathcal{P}_2}(\mathbf{x}))_{left})}{\mathbf{l}!} \cdot (\mathbf{x} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left})^{\mathbf{l}},$$

where we use that by construction we have $C_{\mathcal{P}_2}(\mathbf{x}) = \tilde{C}_{k,i}$ for some $k \in \{1, \dots, M^d\}$. Since

$$\begin{aligned} & \left| \hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) - T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) \right| \\ & \leq \sum_{\substack{\mathbf{l} \in \mathbb{N}_0^d: \\ \|\mathbf{l}\|_1 \leq q}} \frac{\left| (\partial^{\mathbf{l}} \hat{f} - \partial^{\mathbf{l}} f)((C_{\mathcal{P}_2}(\mathbf{x}))_{left}) \right|}{\mathbf{l}!} \cdot |\mathbf{x} - (C_{\mathcal{P}_2}(\mathbf{x}))_{left}|^{\mathbf{l}} \\ (36) \quad & \leq e^d \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^p \end{aligned}$$

the network approximating $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$ is also a good approximation for $T_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$.

As in the proof of Theorem 2 a) we can fomulate the following *four* key steps for the proof of Theorem 2 b):

1. Compute $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$ by recursively defined functions.
2. Approximate the recursive functions by neural networks. The resulting network is a good approximation for $f(\mathbf{x})$ in case that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

3. Approximate the function $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ by deep neural networks.
4. Apply those networks to 2^d slightly shifted partitions of $\mathcal{P}_2$ to approximate $f(\mathbf{x})$ in supremum norm.

A.1.9. *Key step 1: A recursive definition of $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$.* As in the proof of Theorem 2 a) we will use that we can compute $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$ recursively. To do this, we set

$$\begin{aligned}\phi_{1,0} &= (\phi_{1,0}^{(1)}, \dots, \phi_{1,0}^{(d)}) = \mathbf{x} \\ \phi_{2,0} &= (\phi_{2,0}^{(1)}, \dots, \phi_{2,0}^{(d)}) = \mathbf{0}\end{aligned}$$

and

$$\phi_{3,0}^{(\mathbf{l})} = 0 \text{ and } \phi_{4,0}^{(\mathbf{l})} = 0$$

for each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. For $j \in \{1, \dots, M^d\}$ set

$$\phi_{1,j} = \phi_{1,j-1},$$

$$\phi_{2,j} = (C_{j,1})_{left} \cdot \mathbb{1}_{C_{j,1}}(\phi_{1,j-1}) + \phi_{2,j-1},$$

$$\phi_{3,j}^{(\mathbf{l})} = (\partial^{\mathbf{l}} f)((C_{j,1})_{left}) \cdot \mathbb{1}_{C_{j,1}}(\phi_{1,j-1}) + \phi_{3,j-1}^{(\mathbf{l})}$$

and

$$\phi_{4,j}^{(\mathbf{l})} = b_j^{(\mathbf{l})} \cdot \mathbb{1}_{C_{j,1}}(\phi_{1,j-1}) + \phi_{4,j-1}^{(\mathbf{l})}.$$

Furthermore set

$$\phi_{1,M^d+j} = \phi_{1,M^d+j-1}, \quad j \in \{1, \dots, M^d\},$$

$$\phi_{2,M^d+j} = \phi_{2,M^d+j-1} + \tilde{\mathbf{v}}_{j+1},$$

$$\begin{aligned} \phi_{3,M^d+j}^{(\mathbf{l})} = & \sum_{\substack{\mathbf{s} \in \mathbb{N}_0^d \\ \|\mathbf{s}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{\phi_{3,M^d+j-1}^{(\mathbf{l}+\mathbf{s})}}{\mathbf{s}!} \cdot (\tilde{\mathbf{v}}_{j+1})^{\mathbf{s}} \\ & + \left(\lfloor (4 + 2 \cdot \lceil e^d \rceil) \cdot \phi_{4,M^d+j-1}^{(\mathbf{l})} \rfloor - \lceil e^d \rceil - 2 \right) \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1}, \end{aligned}$$

$$\phi_{4,M^d+j}^{(\mathbf{l})} = (4 + 2 \cdot \lceil e^d \rceil) \cdot \phi_{4,M^d+j-1}^{(\mathbf{l})} - \lfloor (4 + 2 \cdot \lceil e^d \rceil) \cdot \phi_{4,M^d+j-1}^{(\mathbf{l})} \rfloor$$

for $j \in \{1, \dots, M^d - 1\}$ and each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and

$$\phi_{5,M^d+j} = \mathbb{1}_{\mathcal{A}^{(j)}}(\phi_{1,M^d+j-1}) \cdot \phi_{2,M^d+j-1} + \phi_{5,M^d+j-1}$$

and

$$\phi_{6,M^d+j}^{(\mathbf{l})} = \mathbb{1}_{\mathcal{A}^{(j)}}(\phi_{1,M^d+j-1}) \cdot \phi_{3,M^d+j-1}^{(\mathbf{l})} + \phi_{6,M^d+j-1}^{(\mathbf{l})}$$

for $j \in \{1, \dots, M^d\}$, where

$$\phi_{5,M^d} = \left(\phi_{5,M^d}^{(1)}, \dots, \phi_{5,M^d}^{(d)} \right) = \mathbf{0}, \quad \phi_{6,M^d}^{(\mathbf{l})} = 0$$

and

$$\begin{aligned} \mathcal{A}^{(j)} = & \left\{ \mathbf{x} \in \mathbb{R}^d : -x^{(k)} + \phi_{2,M^d+j-1}^{(k)} \leq 0 \right. \\ & \left. \text{und } x^{(k)} - \phi_{2,M^d+j-1}^{(k)} - \frac{2a}{M^2} < 0 \text{ for all } k \in \{1, \dots, d\} \right\}. \end{aligned}$$

Finally define

$$\phi_{1,2M^d+1} = \sum_{\substack{\mathbf{l} \in \mathbb{N}_0^d \\ \|\mathbf{l}\|_1 \leq q}} \frac{\phi_{6,2M^d}^{(\mathbf{l})}}{\mathbf{l}!} \cdot (\phi_{1,2M^d} - \phi_{5,2M^d})^{\mathbf{l}}.$$

Our next lemma shows that this recursion computes $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x})$.

LEMMA 11. *Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $s \in (0, 1]$, let $C > 0$ and $\mathbf{x} \in [-a, a]^d$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function and let $\hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}$ be defined as in (35). Define $\phi_{1,2M^d+1}$ recursively as above. Then we have*

$$\phi_{1,2M^d+1} = \hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}).$$

PROOF. Let $\mathbf{x} \in \tilde{C}_{j,i}$ for some $j \in \{1, \dots, M^d\}$, $i \in \{1, \dots, M^d\}$. Then we have $C_{\mathcal{P}_2}(\mathbf{x}) = \tilde{C}_{j,i}$ and $\mathbf{x} \in C_{i,1}$ and it is easy to see that

$$\begin{aligned}\phi_{1,M^d} &= \mathbf{x}, & \phi_{2,M^d} &= (C_{i,1})_{left}, \\ \phi_{3,M^d}^{(\mathbf{l})} &= (\partial^{\mathbf{l}} f)((C_{i,1})_{left}) \text{ and } \phi_{4,M^d}^{(\mathbf{l})} &= b_i^{(\mathbf{l})}\end{aligned}$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. Since

$$\phi_{2,M^d} = (C_{i,1})_{left} = (\tilde{C}_{1,i})_{left}$$

the values of ϕ_{2,M^d+j} successively describe $(\tilde{C}_{j+1,i})_{left}$ according to (34) and we have

$$\mathcal{A}^{(j)} = \tilde{C}_{j,i}.$$

Furthermore we can conclude that

$$\phi_{3,M^d+j}^{(\mathbf{l})} = (\partial^{\mathbf{l}} \hat{f})((\tilde{C}_{j+1,i})_{left}),$$

where we have used that

$$\phi_{4,M^d+j-1}^{(\mathbf{l})} = \sum_{k=1}^{M^d-j} (b_{k+j-1,i}^{(\mathbf{l})} + \lceil e^d \rceil + 2) \cdot (4 + 2\lceil e^d \rceil)^{-k}$$

and

$$\lfloor (4 + 2 \cdot \lceil e^d \rceil) \cdot \phi_{4,1}^{(M^d+j-1)} \rfloor - \lceil e^d \rceil - 2 = b_{j,i}^{(\mathbf{l})}.$$

This leads to

$$\phi_{5,2M^d} = (\tilde{C}_{j,i})_{left}$$

and

$$\phi_{6,2M^d}^{(\mathbf{l})} = (\partial^{\mathbf{l}} \hat{f})((C_{j,i})_{left})$$

for each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and finally implies

$$\phi_{1,2M^d+1} = \hat{T}_{f,q,(C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}).$$

□

A.1.10. *Key step 2: Approximating $\phi_{1,2M^d+1}$ by neural networks.* In this step we show that a neural network approximates $\phi_{1,2M^d+1}$ in case that

$$\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0.$$

We proceed as in the proof of Theorem 2 a) and define a composed neural network, which approximately computes the recursive functions in the definition of $\phi_{1,2M^d+1}$.

LEMMA 12. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $\mathcal{P}_2$ be defined as in (15). Let $p = q + s$ for some $q \in \mathbb{N}_0$ and $s \in (0, 1]$, and let $C > 0$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function. Let $1 \leq a < \infty$. Then there exists for $M \in \mathbb{N}$ sufficiently large (independent of the size of a , but*

$$(37) \quad \begin{aligned} M^{2p} &\geq 2^{4(q+1)+1} \max\{c_{36} \cdot (6 + 2\lceil e^d \rceil)^{4(q+1)}, c_{46} \cdot e^d\} \\ &\quad \cdot \left(\max\left\{a, \|f\|_{C^q([-a, a]^d)}\right\} \right)^{4(q+1)} \end{aligned}$$

must hold), a neural network $\hat{f}_{\text{deep}, \mathcal{P}_2} \in \mathcal{F}(L, r)$ with

$$\begin{aligned} (i) \quad L &= 4M^d + \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil \\ &\quad \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil \\ (ii) \quad r &= \max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot (2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d), \right. \\ &\quad \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} \end{aligned}$$

such that

$$|\hat{f}_{\text{deep}, \mathcal{P}_2}(\mathbf{x}) - f(\mathbf{x})| \leq c_{47} \cdot \left(\max\left\{2a, \|f\|_{C^q([-a, a]^d)}\right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}$$

holds for all $\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0$. The network value is bounded by

$$\begin{aligned} |\hat{f}_{\text{deep}, \mathcal{P}_2}(\mathbf{x})| &\leq 1 + \left(\|f\|_{C^q([-a, a]^d)} \cdot e^{(M^d-1)} \right. \\ &\quad \left. + (4 + 2 \cdot \lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \right) \cdot e^{2ad} \end{aligned}$$

for all $\mathbf{x} \in [-a, a]^d$.

This lemma already shows that a network with depth of order M^d and constant width achieves an approximation error of size $1/M^{2p}$ in case that

$$x \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0.$$

Beside the neural networks already introduced for the proof of Theorem 2 a) we need one further neural network in the construction of $\phi_{1,2M^d+1}$ to finally show [Lemma 12](#).

A further auxiliary neural network. In the following we introduce a network which approximates $\lfloor z \rfloor$ for some $z \in [0, B+1)$ with $B \in \mathbb{N}$. This network helps to compute the values $b_{k,i}^{(1)}$ (see the computation of $\phi_{3,M^d+j}^{(1)}$ and $\phi_{4,M^d+j}^{(1)}$). In the construction of the network we use that

$$\lfloor z \rfloor = \sum_{j=1}^B \mathbb{1}_{[j,\infty)}(z)$$

for $z \in [0, B+1)$ and that each of the indicator functions can be approximated by a neural network

$$\hat{f}_{ind,[j,\infty)}(z) = R \cdot \sigma(z - j) - R \cdot \sigma\left(z - j - \frac{1}{R}\right).$$

LEMMA 13. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $R > 0$, $B \in \mathbb{N}$ and*

$$\hat{f}_{ind,[j,\infty)}(z) = R \cdot \sigma(z - j) - R \cdot \sigma\left(z - j - \frac{1}{R}\right) \in \mathcal{F}(1, 2)$$

for $j \in \{1, \dots, B\}$. Then the neural network

$$\hat{f}_{trunc}(z) = \sum_{j=1}^B \hat{f}_{ind,[j,\infty)}(z) \in \mathcal{F}(1, 2B)$$

satisfies

$$\hat{f}_{trunc}(z) = \lfloor z \rfloor$$

for $z \in [0, B+1)$ and $\min\{|z - j| : j \in \mathbb{N}\} \geq 1/R$.

PROOF. For $z \geq j + 1/R$ ($j \in \{1, \dots, B\}$) we have

$$\hat{f}_{ind,[j,\infty)}(z) = R \cdot (z - j) - R \cdot \left(z - j - \frac{1}{R} \right) = 1 = \mathbb{1}_{[j,\infty)}(z).$$

For $z \leq j$ we have $z - j \leq 0$ and $z - j - 1/R \leq 0$, such that

$$\hat{f}_{ind,[j,\infty)}(z) = 0 = \mathbb{1}_{[j,\infty)}(z).$$

Consequently we have

$$\hat{f}_{ind,[j,\infty)}(z) = \mathbb{1}_{[j,\infty)}(z)$$

in case $\min\{|z - j| : j \in \mathbb{N}\} \geq 1/R$. Since

$$\lfloor z \rfloor = \sum_{j=1}^B \mathbb{1}_{[j,\infty)}(z)$$

for $z \in [0, B + 1)$ this shows the assertion. $\square$

In order to show that $\hat{f}_{trunc}$ computes in our neural network the correct value, we will need the following auxiliary result.

LEMMA 14. *Let $b_{k,i}^{(1)} \in \mathbb{Z}$ such that*

$$|b_{k,i}^{(1)}| \leq e^d + 1.$$

Then

$$\begin{aligned} & \min_{r \in \mathbb{N}} \left| \sum_{k=1}^{M^d-j} (b_{k+j-1,i}^{(1)} + \lceil e^d \rceil + 2) \cdot (4 + 2\lceil e^d \rceil)^{-k+1} - r \right| \\ & \geq \frac{1}{(4 + 2\lceil e^d \rceil)^{M^d-j-1}} \end{aligned}$$

holds for any $j \in \{1, \dots, M^d - 1\}$.

PROOF. Because of

$$1 \leq b_{k,i}^{(1)} + \lceil e^d \rceil + 2 \leq 2\lceil e^d \rceil + 3$$

we have

$$\min_{r \in \mathbb{N}} \left| \sum_{k=1}^{M^d-j} (b_{k+j-1,i}^{(1)} + \lceil e^d \rceil + 2) \cdot (4 + 2\lceil e^d \rceil)^{-k+1} - r \right|$$

$$\begin{aligned}
 &\geq \min \left\{ (4 + 2\lceil e^d \rceil)^{-M^d+j+1}, 1 - \sum_{k=1}^{M^d-j-1} (2\lceil e^d \rceil + 3) \cdot (4 + 2\lceil e^d \rceil)^{-k} \right\} \\
 &= \min \left\{ (4 + 2\lceil e^d \rceil)^{-M^d+j+1}, \sum_{k=M^d-j}^{\infty} (2\lceil e^d \rceil + 3) \cdot (4 + 2\lceil e^d \rceil)^{-k} \right\} \\
 &= \frac{1}{(4 + 2\lceil e^d \rceil)^{M^d-j-1}}
 \end{aligned}$$

for $j \in \{1, \dots, M^d - 1\}$. $\square$

In the proof of [Lemma 12](#) every function of $\phi_{1,2M^d+1}$ is computed by a neural network. In particular, the indicator functions in $\phi_{2,j}$, $\phi_{3,j}^{(1)}$ and $\phi_{4,j}^{(1)}$ ($j \in \{1, \dots, M^d\}$, $\mathbf{l} \in \mathbb{N}_0^d$, $\|\mathbf{l}\|_1 \leq q$) are computed by [Lemma 6 a\)](#), while we apply the identity network to shift the computed values from the previous step. The functions $\phi_{3,M^d+j}^{(1)}$ and $\phi_{4,M^d+j}^{(1)}$ are then computed with the help of [Lemma 11](#), while we again use the identity network to shift values in the next hidden layers. For the functions ϕ_{5,M^d+j} and $\phi_{6,M^d+j}^{(1)}$ we use the network of [Lemma 6 b\)](#) to successively compute $(C_{\mathcal{P}_2}(\mathbf{x}))_{left}$ and the derivatives on the cube $C_{\mathcal{P}_2}(\mathbf{x})$. The final Taylor polynomial in $\phi_{1,2M^d+1}$ is then approximated with the help of [Lemma 5](#).

PROOF OF [LEMMA 12](#). In a *first step of the proof* we describe how the recursively defined function $\phi_{1,2M^d+1}$ of [Lemma 11](#) can be approximated by neural networks. In the construction we will use the network

$$\hat{f}_{ind,[\mathbf{a},\mathbf{b}]} \in \mathcal{F}(2, 2d)$$

of [Lemma 6](#), which approximates the indicator function $\mathbb{1}_{[\mathbf{a},\mathbf{b}]}(\mathbf{x})$ for some $\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$ and $B_M \in \mathbb{N}$ with

$$b^{(i)} - a^{(i)} \geq \frac{2}{B_M} \text{ for all } i \in \{1, \dots, d\}$$

and the network

$$\hat{f}_{test} \in \mathcal{F}(2, 2 \cdot (2d + 2))$$

of [Lemma 6](#), which approximates

$$s \cdot \mathbb{1}_{[\mathbf{a},\mathbf{b}]}(\mathbf{x}).$$

Observe that for $B_M \in \mathbb{N}$ and

$$x^{(i)} \notin \left[a^{(i)}, a^{(i)} + \frac{1}{B_M} \right) \cup \left(b^{(i)} - \frac{1}{B_M}, b^{(i)} \right) \text{ for all } i \in \{1, \dots, d\}$$

we have

$$\hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x})$$

and

$$\hat{f}_{test}(\mathbf{x}, \mathbf{a}, \mathbf{b}, s) = s \cdot \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{x}).$$

Here we treat B_M as R in [Lemma 6](#). For some vector $\mathbf{v} \in \mathbb{R}^d$ it follows

$$\mathbf{v} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) = \left(v^{(1)} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}), \dots, v^{(d)} \cdot \hat{f}_{ind, [\mathbf{a}, \mathbf{b}]}(\mathbf{x}) \right).$$

Furthermore we use the networks

$$\hat{f}_{trunc, i} \in \mathcal{F}(1, 2 \cdot (4 + 2^{\lceil e^d \rceil})), \quad (i \in \{1, \dots, M^d - 1\})$$

of [Lemma 13](#), which satisfies

$$\hat{f}_{trunc, i}(z) = \lfloor z \rfloor$$

for $z \in [0, 5 + 2^{\lceil e^d \rceil})$ with $\min\{|z - j| : j \in \mathbb{N}\} \geq \frac{1}{R_{M, i}}$. Here we choose

$$R = R_{M, i} = \frac{1}{(4 + 2^{\lceil e^d \rceil})^{M^d - i - 1}} \quad \text{and} \quad B = 4 + 2^{\lceil e^d \rceil}$$

in [Lemma 13](#).

To compute the final Taylor polynomial we use the network

$$\hat{f}_p \in \mathcal{F} \left(B_{M, p} \cdot \lceil \log_2(\max\{q + 1, 2\}) \rceil, 18(q + 1) \cdot \binom{d + q}{d} \right)$$

from [Lemma 5](#) satisfying

$$\begin{aligned} & \left| \hat{f}_p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{q}} \right) - p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{q}} \right) \right| \\ & \leq c_{36} \cdot (6 + 2^{\lceil e^d \rceil})^{4(q+1)} \cdot \bar{r}(p) \cdot \left(\max \left\{ \|f\|_{C^q([-a, a]^d)}, a \right\} \right)^{4(q+1)} \\ (38) \quad & \cdot M^{d \cdot 4 \cdot (q+1)} \cdot e^{4(q+1) \cdot (M^d - 1)} \cdot 4^{-B_{M, p}} \end{aligned}$$

for all $z^{(1)}, \dots, z^{(d)}, y_1, \dots, y_{\binom{d+q}{d}}$ contained in

$$\begin{aligned} & \left[-2 \cdot \max \left\{ \|f\|_{C^q([-a,a]^d)}, a \right\} \cdot e^{(M^d-1)} + (4 + 2\lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)}, \right. \\ & \quad \left. 2 \cdot \max \left\{ \|f\|_{C^q([-a,a]^d)}, a \right\} \cdot e^{(M^d-1)} + (4 + 2\lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \right] \\ & \subset \left[-2 \cdot \max \left\{ \|f\|_{C^q([-a,a]^d)}, a \right\} \cdot e^{(M^d-1)} \cdot (6 + 2\lceil e^d \rceil) \cdot M^d, \right. \\ & \quad \left. 2 \cdot \max \left\{ \|f\|_{C^q([-a,a]^d)}, a \right\} \cdot e^{(M^d-1)} \cdot (6 + 2\lceil e^d \rceil) \cdot M^d \right] \end{aligned}$$

where

$$B_{M,p} = \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil$$

satisfies (due to the assumptions on M)

$$\begin{aligned} B_{M,p} \geq \log_4 & \left(2 \cdot 4^{2 \cdot (q+1)} \left(2 \cdot \max \left\{ \|f\|_{C^q([-a,a]^d)}, a \right\} \cdot e^{(M^d-1)} \right. \right. \\ & \quad \left. \left. + (4 + 2\lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \right)^{2 \cdot (q+1)} \right) \end{aligned}$$

Here we treat $B_{M,p}$ as R in [Lemma 5](#). Again we treat a polynomial of degree zero as a polynomial of degree 1, where we choose $r_i = 0$ for all coefficients greater than zero. Thus we substitute $\log_2(q+1)$ by $\log_2(\max\{q+1, 2\})$ in the definition of L in [Lemma 5](#).

To compute $\phi_{1,j}, \phi_{2,j}, \phi_{3,j}^{(1)}$ and $\phi_{4,j}^{(1)}$ for $j \in \{0, \dots, M^d\}$ and each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ we use the networks

$$\begin{aligned} \hat{\phi}_{1,0} &= \left(\hat{\phi}_{1,0}^{(1)}, \dots, \hat{\phi}_{1,0}^{(d)} \right) = \mathbf{x} \\ \hat{\phi}_{2,0} &= \left(\hat{\phi}_{2,0}^{(1)}, \dots, \hat{\phi}_{2,0}^{(d)} \right) = \mathbf{0}, \\ \hat{\phi}_{3,0}^{(1)} &= 0 \text{ and } \hat{\phi}_{4,0}^{(1)} = 0. \end{aligned}$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. For $j \in \{1, \dots, M^d\}$ we set

$$\begin{aligned} \hat{\phi}_{1,j} &= \hat{f}_{id}^2 \left(\hat{\phi}_{1,j-1} \right), \\ \hat{\phi}_{2,j} &= (C_{j,1})_{left} \cdot \hat{f}_{ind,C_{j,1}}(\hat{\phi}_{1,j-1}) + \hat{f}_{id}^2(\hat{\phi}_{2,j-1}), \\ \hat{\phi}_{3,j}^{(1)} &= (\partial^{\mathbf{l}} f)((C_{j,1})_{left}) \cdot \hat{f}_{ind,C_{j,1}}(\hat{\phi}_{1,j-1}) + \hat{f}_{id}^2(\hat{\phi}_{3,j-1}^{(1)}), \end{aligned}$$

$$\hat{\phi}_{4,j}^{(1)} = b_j^{(1)} \cdot \hat{f}_{ind,C_{j,1}}(\hat{\phi}_{1,j-1}) + \hat{f}_{id}^2(\hat{\phi}_{4,j-1}^{(1)})$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. It is easy to see that this parallelized network needs $2M^d$ hidden layers and $2d + d \cdot (2d + 2) + 2 \cdot \binom{d+q}{d} \cdot (2d + 2)$ neurons per layer, where we have used that we have $\binom{d+q}{d}$ different vectors $\mathbf{l} \in \mathbb{N}_0^d$ satisfying $\|\mathbf{l}\|_1 \leq q$.

To compute $\phi_{1,M^d+j}, \phi_{5,M^d+j}$ and $\phi_{6,M^d+j}^{(1)}$ for $j \in \{1, \dots, M^d\}$ and $\phi_{2,M^d+j}, \phi_{3,M^d+j}^{(1)}$ and $\phi_{4,M^d+j}^{(1)}$ for $j \in \{1, \dots, M^d - 1\}$ we use the networks

$$\begin{aligned} \hat{\phi}_{1,M^d+j} &= \hat{f}_{id}^2(\hat{\phi}_{1,M^d+j-1}), \quad j \in \{1, \dots, M^d\}, \\ \hat{\phi}_{2,M^d+j} &= \hat{f}_{id}^2(\hat{\phi}_{2,M^d+j-1} + \tilde{\mathbf{v}}_{j+1}), \\ \hat{\phi}_{3,M^d+j}^{(1)} &= \hat{f}_{id} \left(\hat{f}_{id} \left(\sum_{\substack{\mathbf{s} \in \mathbb{N}_0^d \\ \|\mathbf{s}\|_1 \leq q - \|\mathbf{l}\|_1}} \frac{\hat{\phi}_{3,M^d+j-1}^{(1+\mathbf{s})}}{\mathbf{s}!} \cdot (\tilde{\mathbf{v}}_{j+1})^{\mathbf{s}} \right) \right. \\ &\quad \left. + \left(\hat{f}_{trunc,j} \left((4 + 2 \cdot \lceil e^d \rceil) \cdot \hat{\phi}_{4,M^d+j-1}^{(1)} \right. \right. \right. \\ &\quad \left. \left. \left. - \lceil e^d \rceil - 2 \right) \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p - \|\mathbf{l}\|_1} \right) \right), \\ \hat{\phi}_{4,M^d+j}^{(1)} &= \hat{f}_{id} \left(\hat{f}_{id} \left((4 + 2 \cdot \lceil e^d \rceil) \cdot \hat{\phi}_{4,M^d+j-1}^{(1)} \right) \right. \\ &\quad \left. - \hat{f}_{trunc,j} \left((4 + 2 \cdot \lceil e^d \rceil) \cdot \hat{\phi}_{4,M^d+j-1}^{(1)} \right) \right) \end{aligned}$$

for $j \in \{1, \dots, M^d - 1\}$,

$$\begin{aligned} \hat{\phi}_{5,M^d+j}^{(k)} &= \hat{f}_{test} \left(\hat{\phi}_{1,M^d+j-1}, \hat{\phi}_{2,M^d+j-1}, \right. \\ &\quad \left. \hat{\phi}_{2,M^d+j-1} + \frac{2a}{M^2} \cdot \mathbf{1}, \hat{\phi}_{2,M^d+j-1}^{(k)} \right) \\ (39) \quad &+ \hat{f}_{id}^2 \left(\hat{\phi}_{5,M^d+j-1}^{(k)} \right) \end{aligned}$$

and

$$\begin{aligned} \hat{\phi}_{6,M^d+j}^{(1)} &= \hat{f}_{test} \left(\hat{\phi}_{1,M^d+j-1}, \hat{\phi}_{2,M^d+j-1}, \right. \\ &\quad \left. \hat{\phi}_{2,M^d+j-1} + \frac{2a}{M^2} \cdot \mathbf{1}, \hat{\phi}_{3,M^d+j-1}^{(1)} \right) \\ (40) \quad &+ \hat{f}_{id}^2 \left(\hat{\phi}_{6,M^d+j-1}^{(1)} \right), \end{aligned}$$

where $\hat{\phi}_{5,M^d} = (\hat{\phi}_{5,M^d}^{(1)}, \dots, \hat{\phi}_{5,M^d}^{(d)}) = \mathbf{0}$ and $\hat{\phi}_{6,M^d}^{(\mathbf{l})} = 0$ for each $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. Again it is easy to see, that this parallelized and composed network needs $4M^d$ hidden layers and has width r with

$$\begin{aligned} r &= 2d + 2d + 2 \cdot \binom{d+q}{d} \cdot (2 \cdot (4 + 2\lceil e^d \rceil) + 2) + d \cdot (2 \cdot (2 + 2d) + 2) \\ &\quad + \binom{d+q}{d} \cdot (2 \cdot (2 + 2d) + 2) \\ &= 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot (2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d). \end{aligned}$$

Choose $\mathbf{l}_1, \dots, \mathbf{l}_{\binom{d+q}{d}}$ such that

$$\left\{ \mathbf{l}_1, \dots, \mathbf{l}_{\binom{d+q}{d}} \right\} = \left\{ \mathbf{s} \in \mathbb{N}_0^d : \|\mathbf{s}\|_1 \leq q \right\}$$

holds. The value of $\phi_{1,2M^d+1}$ can then be computed by

$$(41) \quad \hat{\phi}_{1,2M^d+1} = \hat{f}_p(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{d}}),$$

where

$$\mathbf{z} = \hat{\phi}_{1,2M^d} - \hat{\phi}_{5,2M^d}$$

and

$$y_v = \hat{\phi}_{6,2M^d}^{(\mathbf{l}_v)}$$

for $v \in \left\{ 1, \dots, \binom{d+q}{d} \right\}$. The coefficients $r_1, \dots, r_{\binom{d+q}{d}}$ in [Lemma 5](#) are chosen as

$$r_i = \frac{1}{\mathbf{l}_i!}, \quad i \in \left\{ 1, \dots, \binom{d+q}{d} \right\}.$$

The final network $\hat{\phi}_{1,2M^d+1}$ is then contained in the class

$$\mathcal{F}(4M^d + B_{M,p} \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil, r)$$

with

$$\begin{aligned} r &= \max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot (2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d), \right. \\ &\quad \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} \end{aligned}$$

and we set

$$\hat{f}_{deep, \mathcal{P}_2}(\mathbf{x}) = \hat{\phi}_{1, 2M^d+1}.$$

In a *second step of the proof* we analyze the error of the network $\hat{f}_{deep, \mathcal{P}_2}(\mathbf{x})$ in case that

$$B_M \geq M^{2p+2}$$

and

$$(42) \quad \mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2d}\}} (C_{k,2})_{1/M^{2p+2}}^0.$$

Using Lemma 6 it is easy to see that we have $\hat{\phi}_{1,j} = \phi_{1,j}$, $\hat{\phi}_{2,j} = \phi_{2,j}$, $\hat{\phi}_{3,j}^{(1)} = \phi_{3,j}^{(1)}$ and $\hat{\phi}_{4,j}^{(1)} = \phi_{4,j}^{(1)}$ for all $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and all $j \in \{1, \dots, M^d\}$. This implies $\hat{\phi}_{1, M^d+j} = \phi_{1, M^d+j}$ and $\hat{\phi}_{2, M^d+j} = \phi_{2, M^d+j}$ for all $j \in \{1, \dots, M^d\}$. Via induction we can conclude from Lemma 13 and Lemma 14 $\hat{\phi}_{3, M^d+j}^{(1)} = \phi_{3, M^d+j}^{(1)}$ and $\hat{\phi}_{4, M^d+j}^{(1)} = \phi_{4, M^d+j}^{(1)}$ for all $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$ and all $j \in \{1, \dots, M^d\}$, and a second induction implies $\hat{\phi}_{5, M^d+j}^{(1)} = \phi_{5, M^d+j}^{(1)}$ and $\hat{\phi}_{6, M^d+j}^{(1)} = \phi_{6, M^d+j}^{(1)}$ for all $j \in \{1, \dots, M^d\}$.

Thus it follows that

$$\left| \hat{\phi}_{1, 2M^d} - \hat{\phi}_{5, 2M^d} \right| = \left| \mathbf{x} - \phi_{5, 2M^d} \right| \leq 2a$$

and

$$\begin{aligned} \left| \hat{\phi}_{6, 2M^d}^{(1)} \right| &= \left| \phi_{6, 2M^d}^{(1)} \right| = \left| (\partial^{\mathbf{l}} \hat{f})((C_{\mathcal{P}_2}(\mathbf{x}))_{left}) \right| \\ &\leq \left| (\partial^{\mathbf{l}} \hat{f})((C_{\mathcal{P}_2}(\mathbf{x}))_{left}) - (\partial^{\mathbf{l}} f)((C_{\mathcal{P}_2}(\mathbf{x}))_{left}) \right| \\ &\quad + \left| (\partial^{\mathbf{l}} f)((C_{\mathcal{P}_2}(\mathbf{x}))_{left}) \right| \\ &\leq 2 \cdot \max \left\{ 1, \|f\|_{C^q([-a, a]^d)} \right\}, \end{aligned}$$

where we have used that

$$M^2 \geq c_{46}^{1/p-q} \cdot 2a.$$

Therefore the input of $\hat{f}_p$ in (41) is contained in the interval, where (38) holds. We get

$$\left| \hat{f}_{deep, \mathcal{P}_2}(\mathbf{x}) - \hat{T}_{f, q, (C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) \right| = \left| \hat{\phi}_{1, 2M^d+1} - \phi_{1, 2M^d+1} \right|$$

$$\leq c_{36} \cdot (6 + 2\lceil e^d \rceil)^{4(q+1)} \cdot \left(2 \cdot \max \left\{ a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}.$$

This together with [Lemma 1](#) and (36) shows the first assertion of the lemma. Furthermore by [Lemma 1](#) we can bound the value of the network by

$$\begin{aligned} & \left| \hat{f}_{deep, \mathcal{P}_2}(\mathbf{x}) \right| \\ & \leq c_{36} \cdot (6 + 2\lceil e^d \rceil)^{4(q+1)} \cdot \left(2 \cdot \max \left\{ a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}} \\ & \quad + \left| T_{f, q, (C_{\mathcal{P}_2}(\mathbf{x}))_{left}}(\mathbf{x}) - f(\mathbf{x}) \right| + |f(\mathbf{x})| \\ & \leq 2^{4(q+1)+1} \cdot c_{48} \cdot (6 + 2\lceil e^d \rceil)^{4(q+1)} \cdot \left(\max \left\{ a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}} \\ & \quad + |f(\mathbf{x})| \\ & \leq 2 \cdot \max \left\{ \|f\|_{\infty, [-a, a]^d}, 1 \right\}, \end{aligned}$$

in case that (42). Here we have used that

$$M^{2p} > 2^{4(q+1)+1} \cdot c_{48} \cdot (6 + 2\lceil e^d \rceil)^{4(q+1)} \cdot \left(\max \left\{ a, \|f\|_{C^q([-a, a]^d)} \right\} \right)^{4(q+1)}.$$

In a *last step of the proof* we analyze the bound of $\hat{f}_{deep, \mathcal{P}_2}(\mathbf{x})$ in case that

$$\mathbf{x} \in \bigcup_{k \in \{1, \dots, M^{2p}\}} C_{k,2} \setminus (C_{k,2})_{1/M^{2p+2}}^0.$$

Then the networks $\hat{f}_{ind, C_{j,1}}$ ($j \in \{1, \dots, M^d\}$), $\hat{f}_{test}$ and $\hat{f}_{trunc, i}$ ($i \in \{1, \dots, M^d - 1\}$) are not exact (see [Lemma 6](#) and [Lemma 13](#)). According to the definition of $\hat{f}_{trunc, i}$ ($i \in \{1, \dots, M^d - 1\}$) with $B = 4 + 2\lceil e^d \rceil$ in [Lemma 13](#), the value of the network is contained in the interval $[0, 4 + 2\lceil e^d \rceil]$ independent of its input. Therefore we have

$$\left| \hat{f}_{trunc, i}((4 + 2\lceil e^d \rceil) \cdot z) - \lceil e^d \rceil - 2 \right| \leq 4 + 2 \cdot \lceil e^d \rceil \quad z \in \mathbb{R}.$$

For $\mathbf{x} \in C_{i,1}$ ($i \in \{1, \dots, M^d\}$) we can conclude that

$$-a \leq \hat{\phi}_{2, M^d}^{(k)} \leq a - \frac{2a}{M}, \quad k \in \{1, \dots, d\}.$$

Here we have used that $\hat{f}_{ind, C_{i,1}}(\mathbf{x}) \in [0, 1]$ and $-a \leq (C_{i,1})_{left}^{(k)} \leq a - 2a/M$. Analogous we can conclude

$$\left| \hat{\phi}_{3, M^d}^{(1)} \right| \leq |(\partial^1 f)((C_{i,1})_{left})|$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. Using those bounds we can conclude that

$$\begin{aligned} \left| \hat{\phi}_{3,M^d+1}^{(\mathbf{l})} \right| &\leq \sum_{s=0}^{\infty} \frac{\|f\|_{C^q([-a,a]^d)}}{s!} \cdot a^s + (4 + 2\lceil e^d \rceil) \cdot c_{46} \cdot \left(\frac{2a}{M^2} \right)^{p-\|\mathbf{l}\|_1} \\ &\leq e^a \cdot \|f\|_{C^q([-a,a]^d)} + (4 + 2\lceil e^d \rceil), \end{aligned}$$

where we have used that

$$M^2 \geq c_{46}^{\frac{1}{p-q}} \cdot 2a.$$

Then it can be shown by induction that

$$\left| \hat{\phi}_{3,M^d+j}^{(\mathbf{l})} \right| \leq \|f\|_{C^q([-a,a]^d)} \cdot e^{j \cdot 2ad/M^2} + (4 + 2 \cdot \lceil e^d \rceil) \cdot \sum_{k=0}^{j-1} e^{k \cdot 2ad/M^2}.$$

Furthermore we have

$$|\hat{\phi}_{2,2M^d}| \leq a.$$

Since $\hat{f}_{test}$ produces for at most one j ($j \in \{1, \dots, M^d\}$) in (39) and (40) a value not equal to zero, this leads to

$$\left| \hat{\phi}_{5,2M^d}^{(k)} \right| \leq a \quad (k \in \{1, \dots, d\})$$

and

$$\begin{aligned} \left| \hat{\phi}_{6,2M^d}^{(\mathbf{l})} \right| &\leq \|f\|_{C^q([-a,a]^d)} \cdot e^{(M^d-1) \cdot 2ad/M^2} + (4 + 2 \cdot \lceil e^d \rceil) \cdot \sum_{k=0}^{M^d-2} e^{k \cdot 2ad/M^2} \\ &\leq \|f\|_{C^q([-a,a]^d)} \cdot e^{(M^d-1)} + (4 + 2 \cdot \lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \end{aligned}$$

for $\mathbf{l} \in \mathbb{N}_0^d$ with $\|\mathbf{l}\|_1 \leq q$. Due to the fact that all components are contained in the interval, where (38) holds, we can bound the value of $\hat{f}_{deep, \mathcal{P}_2}(\mathbf{x})$ by

$$\begin{aligned} \left| \hat{f}_{deep, \mathcal{P}_2}(\mathbf{x}) \right| &\leq \left| \hat{f}_p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{d}} \right) - p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{d}} \right) \right| \\ &\quad + \left| p \left(\mathbf{z}, y_1, \dots, y_{\binom{d+q}{d}} \right) \right| \\ &\leq 1 + \left| \sum_{\substack{\mathbf{l} \in \mathbb{N}_0^d \\ \|\mathbf{l}\|_1 \leq q}} \frac{1}{\mathbf{l}!} \cdot \left(\|f\|_{C^q([-a,a]^d)} \cdot e^{(M^d-1)} \right) \right| \end{aligned}$$

$$\begin{aligned}
& + (4 + 2 \cdot \lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \cdot (2a)^{\|\mathbf{l}\|_1} \Bigg) \Bigg| \\
& \leq 1 + \left| \left(\|f\|_{C^q([-a,a]^d)} \cdot e^{(M^d-1)} \right. \right. \\
& \quad \left. \left. + (4 + 2 \cdot \lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \right) \cdot e^{2ad} \right|.
\end{aligned}$$

This shows the assertion of the lemma. $\square$

A.1.11. *Key step 3: Approximating $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ by deep neural networks.* In order to approximate $f(\mathbf{x})$ in supremum norm, we further approximate the function $w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x})$ by a neural network, where $w_{\mathcal{P}_2}(\mathbf{x})$ is defined as in (22). The result is the following.

LEMMA 15. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$ and $M \in \mathbb{N}_0$ sufficiently large (independent of the size of a , but*

$$\begin{aligned}
M^{2p} \geq & 2^{4(q+1)} \cdot \max\{c_{36}(6 + 2\lceil e^d \rceil)^{4(q+1)}, c_{46} \cdot e^d\} \\
& \cdot \left(\max\left\{a, \|f\|_{C^q([-a,a]^d)}\right\} \right)^{4(q+1)}
\end{aligned}$$

must hold). Let $p = q + s$ for some $q \in \mathbb{N}_0$, $s \in (0, 1]$ and let $C > 0$. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a (p, C) -smooth function and let $w_{\mathcal{P}_2}$ be defined as in (22). Then there exists a network

$$\hat{f} \in \mathcal{F}(L, r)$$

with

$$\begin{aligned}
L = & 5M^d + \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil \cdot \lceil \log_2(\max\{q, d\} + 1) \rceil \\
& + \lceil \log_4(M^{2p}) \rceil
\end{aligned}$$

and

$$\begin{aligned}
r = & \max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot \left(2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d \right), \right. \\
& \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} + 6d^2 + 20d + 2.
\end{aligned}$$

such that

$$\left| \hat{f}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \cdot f(\mathbf{x}) \right| \leq c_{49} \cdot \left(\max\left\{2a, \|f\|_{C^q([-a,a]^d)}\right\} \right)^{4(q+1)} \cdot \frac{1}{M^{2p}}$$

for $\mathbf{x} \in [-a, a]^d$.

As in the proof of Theorem 2 a) we need some further auxiliary lemmata to show this result. First we show that each weight $w_{\mathcal{P}_2}(\mathbf{x})$ can also be approximated by a very deep neural network. Here we use the same construction as described in Lemma 9 with the only difference that $(C_{\mathcal{P}_2}(\mathbf{x}))_{left}$ is computed with the help of $\hat{\phi}_{5,2M^d}$ of Lemma 12.

LEMMA 16. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$ and $M \geq 4^{4d+1} \cdot d$. Let $\mathcal{P}_2$ be the partition defined in (15) and let $w_{\mathcal{P}_2}(\mathbf{x})$ be defined by (22). Then there exists a neural network*

$$\hat{f}_{w_{\mathcal{P}_2}, deep} \in \mathcal{F} \left(4M^d + 1 + \lceil \log_4(M^{2p}) \rceil \cdot \lceil \log_2(d) \rceil, r \right),$$

with

$$r = \max \{18d, 4d^2 + 10d\}$$

such that

$$\left| \hat{f}_{w_{\mathcal{P}_2}, deep}(\mathbf{x}) - w_{\mathcal{P}_2}(\mathbf{x}) \right| \leq 4^{4d+1} \cdot d \cdot \frac{1}{M^{2p}}$$

for $\mathbf{x} \in \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0$ and

$$|\hat{f}_{w_{\mathcal{P}_2}, deep}(\mathbf{x})| \leq 2$$

for $\mathbf{x} \in [-a, a]^d$.

PROOF. The result follows directly from the proof of Lemma 9 with the only difference that we use $\hat{\phi}_{5,2M^d}$ of Lemma 12 (with $4M^d$ hidden layers and $d \cdot (2 \cdot (2d+2) + 2) + 2d$ neurons per layer) and $\hat{f}_{id}^{2M^d}$ to compute the value of $(C_{\mathcal{P}_2}(\mathbf{x}))_{left}$ and to shift the value of $\mathbf{x}$ in the next layer, respectively. $\square$

As Lemma 10 the following lemma checks whether the input $\mathbf{x}$ lies with at least a distance of $1/M^{2p+2}$ away from the boundaries of the cubes of the partition $\mathcal{P}_1$ and $\mathcal{P}_2$ or not. The construction is similar to the one in Lemma 10 with the main difference that we check successively if $\mathbf{x}$ is contained on the boundaries of the cube or not. In particular, the function

$$f_1(\mathbf{x}) = 1 - \sum_{i \in \{1, \dots, M^d\}} \mathbf{1}_{(C_{i,1})_{1/M^{2p+2}}^0}(\mathbf{x})$$

is computed within $2M^d$ layers by successively applying the networks $\hat{f}_{ind, (C_{j,1})_{1/M^{2p+2}}^0}^0$ ($j \in \{1, \dots, M^d\}$) in consecutive layers and by shifting the

value of the previous layer with the identity network. Analogously we compute

$$\mathbb{1}_{\bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

by successively checking whether $\mathbf{x}$ lies on the boundaries of a cube $\tilde{C}_{j,i}$ ($j \in \{1, \dots, M^d\}$) or not. Here we apply successively the network of [Lemma 6 b](#)).

LEMMA 17. *Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the ReLU activation function $\sigma(x) = \max\{x, 0\}$. Let $1 \leq a < \infty$. Let $C_{i,2}$ ($i \in \{1, \dots, M^{2d}\}$) be the cubes of partition $\mathcal{P}_2$ as described in (15) and let $M \in \mathbb{N}$. Then there exists a neural network*

$$\hat{f}_{check, deep, \mathcal{P}_2}(\mathbf{x}) \in \mathcal{F}(5M^d, 2d^2 + 6d + 2)$$

satisfying

$$\hat{f}_{check, deep, \mathcal{P}_2}(\mathbf{x}) = \mathbb{1}_{\bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})_{1/M^{2p+2}}^0}(\mathbf{x})$$

for $\mathbf{x} \notin \bigcup_{i \in \{1, \dots, M^{2d}\}} (C_{i,2})_{1/M^{2p+2}}^0 \setminus (C_{i,2})_{2/M^{2p+2}}^0$ and

$$\hat{f}_{check, deep, \mathcal{P}_2}(\mathbf{x}) \in [0, 1]$$

for $\mathbf{x} \in [-a, a]^d$.

PROOF. The result follows by a straightforward modification of the proof of [Lemma 10](#). Here the value of $(C_{\mathcal{P}_1}(\mathbf{x}))_{left}$ is computed by the network $\hat{\phi}_{2, M^d}$ of [Lemma 12](#) with $2M^d$ hidden layers and $d \cdot (2d+2)$ neurons per layer and $\mathbf{x}$ is shifted in consecutive layers by successively applying $\hat{f}_{id} \in \mathcal{F}(1, 2)$. Furthermore we compute

$$f_1(\mathbf{x}) = 1 - \sum_{i \in \{1, \dots, M^d\}} \mathbb{1}_{(C_{i,1})_{1/M^{2p+2}}^0}(\mathbf{x})$$

by a network

$$\hat{f}_{1,j}(\mathbf{x}) = \hat{f}_{id}^2(\hat{f}_{1,j-1}) - \hat{f}_{ind, (C_{j,1})_{1/M^{2p+2}}^0}(\hat{f}_{id}^{2(j-1)}(\mathbf{x})), \quad j \in \{1, \dots, M^d\},$$

contained in the network class $\mathcal{F}(2j, 2+2d)$, where $\hat{f}_{1,0} = 1$. Next we define

$$\hat{\phi}_{2, M^d+j} = \hat{f}_{id}^3(\hat{\phi}_{2, M^d+j-1} + \tilde{\mathbf{v}}_{j+1}) \in \mathcal{F}(2M^d + 3j, 2d)$$

for $j \in \{1, \dots, M^d\}$. The value of

$$\mathbb{1}_{\bigcup_{i \in \{1, \dots, M^{2d}\}} C_{i,2} \setminus (C_{i,2})^0_{1/M^{2p+2}}}(\mathbf{x})$$

is then successively computed by

$$\begin{aligned} & \hat{f}_{1,M^d+j}(\mathbf{x}) \\ &= 1 - \sigma \left(1 - \hat{f}_{test} \left(\hat{f}_{id}^{2M^d+3(j-1)}(\mathbf{x}), \hat{\phi}_{2,M^d+j-1} + \tilde{\mathbf{v}}_j + \frac{1}{M^{2p+2}} \cdot \mathbf{1}, \right. \right. \\ & \quad \left. \left. \hat{\phi}_{2,M^d+j-1} + \tilde{\mathbf{v}}_j + \frac{2a}{M^2} \cdot \mathbf{1} - \frac{1}{M^{2p+2}} \cdot \mathbf{1}, 1 \right) - \hat{f}_{id}^2 \left(\hat{f}_{1,M^d+j-1} \right) \right) \end{aligned}$$

for $j \in \{1, \dots, M^d\}$, where we use the same idea for the construction as in [Lemma 10](#) but use a *deep* instead of a *wide* network architecture (which means that we successively compute the networks $\hat{f}_{test}$). This network is contained in the network class $\mathcal{F}(2M^d + 3j, 2d + 4)$. Finally we set

$$\hat{f}_{check,deep,\mathcal{P}_2}(\mathbf{x}) = \hat{f}_{1,2M^d}(\mathbf{x})$$

and it is easy to see that this network is contained in

$$\mathcal{F}(5M^d, r)$$

with

$$r = \max\{d \cdot (2d + 2) + 2d + 2 + 2d, 2d + 2d + 2d + 4\} = 2d^2 + 6d + 2$$

and satisfies the assertions of the lemma. $\square$

In the proof of [Lemma 17](#) we use [Lemma 16](#) to approximate $w_{\mathcal{P}_2}(\mathbf{x})$ and [Lemma 12](#) to compute $f(\mathbf{x})$. As in [Lemma 7](#) we apply a network, that *checks* whether $\mathbf{x}$ is close to the boundaries of the cubes of the partition. Thus we define a network $\hat{f}_{\mathcal{P}_2,true}$ (analogously to [Lemma 7](#)), where we use this time the network $\hat{f}_{check,deep,\mathcal{P}_2}$ of [Lemma 17](#).

PROOF OF LEMMA 17. This result follows by a straightforward modification of the proof of [Lemma 7](#). Here we use the network $\hat{f}_{deep,\mathcal{P}_2}$ of [Lemma 12](#) and $\hat{f}_{check,deep,\mathcal{P}_2}$ of [Lemma 17](#) to define

$$\begin{aligned} \hat{f}_{\mathcal{P}_2,true}(\mathbf{x}) &= \sigma \left(\hat{f}_{deep,\mathcal{P}_2}(\mathbf{x}) - B_{true} \cdot \hat{f}_{check,deep,\mathcal{P}_2}(\mathbf{x}) \right) \\ &\quad - \sigma \left(-\hat{f}_{deep,\mathcal{P}_2}(\mathbf{x}) - B_{true} \cdot \hat{f}_{check,deep,\mathcal{P}_2}(\mathbf{x}) \right) \end{aligned}$$

with

$$B_{true} = 1 + \left| \left(\|f\|_{C^q([-a,a]^d)} \cdot e^{(M^d-1)} + (4 + 2 \cdot \lceil e^d \rceil) \cdot (M^d - 1) \cdot e^{(M^d-2)} \right) \cdot e^{2ad} \right|.$$

Remark that by successively applying $\hat{f}_{id}$ to the output of the networks $\hat{f}_{deep, \mathcal{P}_2}$ and $\hat{f}_{check, deep, \mathcal{P}_2}$ we can achieve that both networks have depth

$$5M^d + \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil \cdot \lceil \log_2(\max\{q+1, 2\}) \rceil.$$

Furthermore it is easy to see that this networks needs at most

$$\max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot \left(2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d \right), \right. \\ \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} + 2d^2 + 6d + 2$$

neurons per layer. In the definition of the final network we use the network $\hat{f}_{w_{\mathcal{P}_2}, deep}$ of [Lemma 16](#) and the network $\hat{f}_{mult}$ defined as in the proof of [Lemma 7](#). Again we synchronize the depth of $\hat{f}_{w_{\mathcal{P}_2}, deep}$ and $\hat{f}_{\mathcal{P}_2, true}$ to achieve that both networks have

$$5M^d + \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil \cdot \lceil \log_2(\max\{q, d\} + 1) \rceil$$

many layers. The final network is given by

$$\hat{f}(\mathbf{x}) = \hat{f}_{mult} \left(\hat{f}_{\mathcal{P}_2, true}(\mathbf{x}), \hat{f}_{w_{\mathcal{P}_2}, deep}(\mathbf{x}) \right).$$

This network is contained in the network class $\mathcal{F}(L, r)$ with

$$L = 5M^d + \left\lceil \log_4 \left(M^{2p+4 \cdot d \cdot (q+1)} \cdot e^{4 \cdot (q+1) \cdot (M^d-1)} \right) \right\rceil \cdot \lceil \log_2(\max\{q, d\} + 1) \rceil \\ + \lceil \log_4(M^{2p}) \rceil$$

and

$$r = \max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot \left(2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d \right), \right. \\ \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} + 2d^2 + 6d + 2 + \max\{18d, 4d^2 + 10d\}.$$

$$\leq \max \left\{ 10d + 4d^2 + 2 \cdot \binom{d+q}{d} \cdot \left(2 \cdot (4 + 2\lceil e^d \rceil) + 5 + 2d \right), \right. \\ \left. 18 \cdot (q+1) \cdot \binom{d+q}{d} \right\} + 6d^2 + 20d + 2.$$

With the same argumentation as in the proof of [Lemma 7](#) we can show the assertion. $\square$

A.1.12. *Key step 4: Applying $\hat{f}$ to slightly shifted partitions.* In the proof of Theorem 2 b) our network follows the same construction as in the proof of Theorem 2 a) with the only difference that we use $\hat{f}_1, \dots, \hat{f}_{2^d}$ of [Lemma 17](#).

PROOF OF THEOREM 2 B). The proof follows directly by the proof of Theorem 2 a) with the only difference that we use the deep networks

$$\hat{f}_1, \dots, \hat{f}_{2^d}$$

of [Lemma 17](#) corresponding to the partitions $\mathcal{P}_{1,v}$ and $\mathcal{P}_{2,v}$ ($v \in \{1, \dots, 2^d\}$). $\square$

A.2. *Network accuracy of t_1 and t_2 in Theorem 3.* In this section we present the two induction proofs, which show the approximation errors of the networks t_1 and t_2 of Theorem 3.

PROOF OF (11). We define

$$g_{\max} := \max \left\{ \max_{\substack{i \in \{1, \dots, l\} \\ j \in \{1, \dots, \tilde{N}_i\}}} \|g_j^{(i)}\|_{\infty}, 1 \right\}.$$

Since each $g_j^{(i)}$ satisfies the assumptions of Theorem 2, we can conclude that

$$(43) \quad \left| \hat{f}_{\text{wide}, g_j^{(i)}}(\mathbf{x}) - g_j^{(i)}(\mathbf{x}) \right| \leq c_{50} \cdot a^{4 \cdot (p_{\max} + 1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}$$

for $\mathbf{x} \in [-2 \max\{g_{\max}, a\}, 2 \max\{g_{\max}, a\}]^{K_j^{(i)}}$, where

$$c_{50} \geq c_{11} (2 \cdot g_{\max} \cdot \max\{c_{20}, 1\})^{4 \cdot (p_{\max} + 1)}.$$

We show by induction that

$$(44) \quad \left| \hat{h}_j^{(i)}(\mathbf{x}) - h_j^{(i)}(\mathbf{x}) \right| \leq c_{50} \cdot i \cdot (K_{\max} \cdot C_{\text{Lip}})^{i-1} \cdot a^{4 \cdot (p_{\max} + 1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}.$$

By (43) we can conclude that

$$\left| \hat{h}_j^{(1)}(\mathbf{x}) - h_j^{(1)}(\mathbf{x}) \right| \leq c_{50} \cdot 1 \cdot (K_{\max} \cdot C_{Lip})^{1-1} \cdot a^{4 \cdot (p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}$$

for $j \in \{1, \dots, \tilde{N}_1\}$. Thus we have shown that (44) holds for $i = 1$. Assume now that (44) holds for some $i - 1$ and every $j \in \{1, \dots, \tilde{N}_{i-1}\}$. Then

$$\left| \hat{h}_j^{(i-1)}(\mathbf{x}) \right| \leq \left| \hat{h}_j^{(i-1)}(\mathbf{x}) - h_j^{(i-1)}(\mathbf{x}) \right| + g_{\max} \leq 2 \cdot g_{\max}$$

follows directly by the induction hypothesis. Using (43) and the Lipschitz continuity of $g_j^{(i)}$ we can conclude that

$$\begin{aligned} & \left| \hat{h}_j^{(i)}(\mathbf{x}) - h_j^{(i)}(\mathbf{x}) \right| \\ & \leq \left| \hat{f}_{wide, g_j^{(i)}} \left(\hat{h}_{\sum_{t=1}^{j-1} K_t^{(i)}+1}^{(i-1)}, \dots, \hat{h}_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)} \right) \right. \\ & \quad \left. - g_j^{(i)} \left(\hat{h}_{\sum_{t=1}^{j-1} K_t^{(i)}+1}^{(i-1)}, \dots, \hat{h}_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)} \right) \right| \\ & + \left| g_j^{(i)} \left(\hat{h}_{\sum_{t=1}^{j-1} K_t^{(i)}+1}^{(i-1)}, \dots, \hat{h}_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)} \right) \right. \\ & \quad \left. - g_j^{(i)} \left(h_{\sum_{t=1}^{j-1} K_t^{(i)}+1}^{(i-1)}(x), \dots, h_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}(x) \right) \right| \\ & \leq c_{50} \cdot a^{4 \cdot (p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}} \\ & \quad + K_j^{(i)} \cdot C_{Lip} \cdot c_{50} \cdot (i-1) \cdot (K_{\max} \cdot C_{Lip})^{i-2} \cdot a^{4 \cdot (p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}} \\ & \leq c_{50} \cdot i \cdot (K_{\max} \cdot C_{Lip})^{i-1} \cdot a^{4 \cdot (p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}. \end{aligned}$$

Thus we have shown that there exists a network $t_1(\mathbf{x})$ satisfying

$$\|t_1 - m\|_{\infty, [-a, a]^d} \leq c_{51} \cdot a^{4 \cdot (p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}.$$

□

PROOF OF (14). We define

$$g_{\max} := \max \left\{ \max_{\substack{i \in \{1, \dots, l\} \\ j \in \{1, \dots, \tilde{N}_i\}}} \|g_j^{(i)}\|_{\infty}, 1 \right\}.$$

Since each $g_j^{(i)}$ satisfies the assumptions of m in Theorem 2, for each network $\hat{f}_{deep, g_j^{(i)}}$ the condition

$$(45) \quad |\hat{f}_{g_j^{(i)}}(\mathbf{x}) - g_j^{(i)}(\mathbf{x})| \leq \frac{c_{52}}{(K_{\max} C_{Lip})^l} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}$$

holds for all $x \in [-2 \max\{g_{\max}, a\}, 2 \max\{g_{\max}, a\}]^{K_j^{(i)}}$, where

$$c_{52} = c_{11} \cdot c_{20}^{4 \cdot (p_{\max}+1)} \cdot (2 \cdot g_{\max})^{4 \cdot (p_{\max}+1)} \cdot (K_{\max} \cdot C_{Lip})^l.$$

In the following we show by induction that

$$(46) \quad \left| h_{N_j^{(i)}}(\mathbf{x}) - \hat{h}_{N_j^{(i)}}(\mathbf{x}) \right| \leq \frac{2^i \cdot c_{52}}{(K_{\max} C_{Lip})^{l-i}} \max_{j,i} M_{j,i}^{-2p_j^{(i)}}$$

holds $i \in \{1, \dots, l\}$ and every $j \in \{1, \dots, \tilde{N}_i\}$. Since each network $\hat{f}_{id}$ shifts the values of the input in the next hidden layers without an error we can conclude by (45) that

$$\begin{aligned} & \left| \hat{h}_i(\mathbf{x}) - h_i(\mathbf{x}) \right| \\ &= \left| \hat{f}_{deep, g_i^{(1)}} \left(x^{(\pi(\sum_{t=1}^{j-1} K_t^{(1)}+1))}, \dots, x^{(\pi(\sum_{t=1}^j K_t^{(1)}))} \right) \right. \\ & \quad \left. - g_i^{(1)} \left(x^{(\pi(\sum_{t=1}^{j-1} K_t^{(1)}+1))}, \dots, x^{(\pi(\sum_{t=1}^j K_t^{(1)}))} \right) \right| \\ &\leq \frac{c_{52}}{(K_{\max} C_{Lip})^l} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}} \end{aligned}$$

for $i \in \{1, \dots, \tilde{N}_1\}$. For $\min_{j,i} M_{j,i}^{2p_j^{(i)}} > c_{52} \cdot a^{4(p_{\max}+1)} / (K_{\max} C_{Lip})^l$ we can bound the value of each network by

$$\left| \hat{h}_i(\mathbf{x}) \right| \leq \left| \hat{h}_i(\mathbf{x}) - h_i(\mathbf{x}) \right| + g_{\max} \leq 2 \cdot g_{\max}.$$

Thus we have shown that (46) holds for $i = 1$ and that the output of each $h_i(\mathbf{x})$ ($i \in \{1, \dots, \tilde{N}_1\}$) is contained in the interval, where inequality (45) holds. Assume now that (46) holds for some $i-1$ and every $j \in \{1, \dots, \tilde{N}_{i-1}\}$. Then

$$\left| \hat{h}_{N_j^{(i-1)}}(\mathbf{x}) \right| \leq \left| \hat{h}_{N_j^{(i-1)}}(\mathbf{x}) - h_{N_j^{(i-1)}}(\mathbf{x}) \right| + g_{\max} \leq 2 \cdot g_{\max}$$

follows directly by the induction hypothesis. This together with the induction hypothesis and the Lipschitz continuity of $g_j^{(i)}$ implies for $i \in \{2, \dots, l\}$ and $j \in \{1, \dots, \tilde{N}_i\}$

$$\begin{aligned}
& \left| \hat{h}_{N_j^{(i)}}(\mathbf{x}) - h_{N_j^{(i)}}(\mathbf{x}) \right| \\
&= \left| \hat{f}_{deep, g_j^{(i)}} \left(\hat{h}_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, \hat{h}_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right. \\
&\quad \left. - g_j^{(i)} \left(h_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, h_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right| \\
&\leq \left| \hat{f}_{deep, g_j^{(i)}} \left(\hat{h}_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, \hat{h}_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right. \\
&\quad \left. - g_j^{(i)} \left(\hat{h}_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, \hat{h}_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right| \\
&\quad + \left| g_j^{(i)} \left(\hat{h}_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, \hat{h}_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right. \\
&\quad \left. - g_j^{(i)} \left(h_{N_{\sum_{t=1}^{j-1} K_t^{(i)} + 1}^{(i-1)}}(\mathbf{x}), \dots, h_{N_{\sum_{t=1}^j K_t^{(i)}}^{(i-1)}}(\mathbf{x}) \right) \right| \\
&\leq \frac{c_{52}}{(K_{\max} C_{Lip})^l} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}} \\
&\quad + K_j^{(i)} \cdot C_{Lip} \cdot \frac{2^{i-1} c_{52}}{(K_{\max} C_{Lip})^{l-i+1}} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}} \\
&\leq \frac{2^i \cdot c_{52}}{(K_{\max} \cdot C_{Lip})^{l-i}} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}.
\end{aligned}$$

Thus we have shown that there exists a network $t_2(\mathbf{x})$ satisfying

$$\|t_2 - m\|_{\infty, [-a, a]^d} \leq c_{52} \cdot a^{4(p_{\max}+1)} \cdot \max_{j,i} M_{j,i}^{-2p_j^{(i)}}.$$

□

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